

Technical Manual

for the Installation, Operation and Maintenance of the

FWC12

Foodwaste Collection System

Serial No: 19P006-900-010-01

Plant Model: FWC12

Project No: 19P006

Date of Issue: 20/06/2024 -Prelim

Document number: 19P006-500-010-00

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1 General

1.1 Explanation of Documentation

This documentation is designed to assist with becoming familiar with the system and how to operate it for its intended purposes. Important safety and hazard notices help you operate the system safely and properly. This system documentation is subject to revision. All previous releases are invalid when the new release has been delivered.

1.2 Other Applicable Documents

The components of other manufacturers which are used within the plant (e.g. e-motors), have a risk assessment from the respective manufacturer. The obligation to ensure that their design complies with the applicable national and European regulations is certificated by these manufacturers.

The operation, maintenance, and repair manuals of the other manufacturers for the relevant parts are inseparable components of this technical manual. The instructions contained in these documents regarding safety, set-up, installation, operation, maintenance, dismantling and disposal of the components are to be followed without limitation by the personnel operating the plant.

1.3 Definitions

DFW	De-watered Food Waste Liquor
FOG	Fat Oil & Grease
GWG	Galley Grey Water (sinks, floor drains etc.)

1.4 Abbreviations

°C	Celsius
DIN	German Industry Standard
DN	Diameter Nominal
EN	European Standard
ISO	International Organization for Standardization
kg	Kilogram
kN	Kilo Newton
PPE	Personal Protection Equipment

1.5 Symbols List

Important safety-related notes in this technical manual are signified by symbols. These notes regarding work safety must be adhered to and complied with to secure the protection of persons and the safe operation of the plant.

Note: This symbol indicates and identifies tips and information to help ensure efficient and trouble free operation of the plant.



CAUTION: This symbol indicates and identifies information that, if not complied with, may lead to minor damage or injury.



WARNING: This symbol indicates and identifies information that, if not complied with, may cause death or serious injury.



DANGER: This symbol indicates and identifies information that, if not complied with, will cause death or serious injury.



WARNING: Electrical hazard! This symbol identifies information that, if not complied with, could lead to health hazards, injury or death.



WARNING: Biohazard, This symbol identifies information that, if not complied with, could lead to illness, or death.



DANGER: This symbol indicates and identifies information that, if not complied with, will cause death or serious injury.

1.6 Copyright Protection

The technical manual must be treated confidential. It is distributed exclusively for those persons working on and with the plant. All content-related information, text, drawings, pictures, and other representations are protected as defined by copyright protection law and are subject to other industrial property rights. Any improper utilization is liable to prosecution.

It is not permitted to pass on the technical manual to third parties, to copy it in any way or form - even extracts - or to recycle and/or communicate the contents without written permission of FMD (the manufacturer).

Violations incur the obligation to pay compensation, all other rights reserved. FMD reserves the right to exercise industrial property rights.

1.7 Modifications

Alterations or changes to the treatment plants are only permissible with express FMD written confirmation thereof.

1.8 Liability and Warranty

This technical manual has been compiled while taking into consideration the applicable regulations, current technology and the experiences and developments of many years. FMD assumes no liability for any damages or malfunctions that occur from non-compliance to the contents of the technical manual. The English version of this technical manual is the authorized version; any translations of the manual were created to the best of our ability, however we cannot assume liability for any translation errors.

All obligations of the manufacturer are defined in the purchase contract, which also includes the entire and sole valid warranty agreement. The contents of the technical manual will not amplify or restrict any contractual warranty.

The texts and illustrations do not necessarily correspond exactly to the scope of delivery. The drawings and diagrams are not scaled 1:1. FMD reserves the right to make technical changes to the product within the context of improving the functionality and further development.

2 Safety

This section includes instructions that must be observed during installation, operation, and maintenance of the equipment.

The operation instructions must be read by any contractor, operator or person prior to installation, start-up, operation, or maintenance and must always be available.

2.1 General Operating Information

Personnel charged with installation, operation, maintenance, inspection, and assembly must be appropriately qualified. Before carrying out any work which involves complete or partial opening of the plant, the person responsible must ensure that power to the plant is locked out (following the ships' isolation procedures) and that it is impossible for the plant to be started. Guards and covers may be removed only when the plant has been secured and locked out. All electrical components must be disconnected before disassembly.

Before restarting the plant, all guards must be in place.

2.2 Operator Safety

- Do not operate the equipment without the proper instructions as given within this manual.
- **WHEN IN DOUBT ASK**
- Remember that the equipment operates automatically and can start and stop at any time.
- Isolate any equipment before working on it, or before asking others to work on it.
- When opening a component of the system such as a pump, pipe or housing **WEAR PPE** including, but not limited to, protective clothing, gloves and a face shield.
- **DO NOT ASK** a tradesman to repair the equipment if they may come in contact with the process stream by doing so.
- **MAKE IT SAFE FIRST** flush out bio-residues from the pumps and pipelines and isolate the equipment. Many pipes contain wastewater even when shut down for some time.
- Always electrically isolate the equipment and lock out the local isolator **BEFORE** you dismantle any electrical equipment. Remember that equipment can be turned on by accident.
- Do not undertake any works unless the consequences are carefully thought through.

2.3 Maintenance Safety

It is in your best interest to clean the FWC plant prior to working on it to minimize the danger of infection.



WARNING: Biohazard.

2.4 Proper Use

The FWC12 may only be operated within the performance parameters specified.

The FWC12 may only be used in conjunction with auxiliary equipment that is recommended and approved by FMD.

For safety and functionality reasons the FWC12 may not be altered or modified. If any modification of the system is required, FMD is to be consulted for prior written approval.

The FWC12 is designed and manufactured with the latest technology and is safe to operate.

Any change in the plant which indicates a malfunction (increased noise level, higher temperatures than normal etc.) must be immediately investigated and where possible the fault rectified.

The operator/operating company are required to immediately report to FMD all changes that may or will detrimentally affect the condition and safety of the system. This also applies to warranty and warranty claims.

2.5 Qualified Personnel

All personnel that are instructed to work with or on the system must observe the rules and regulations for operational safety and accident prevention and must have read the system documentation, including all warning notices, prior to starting work. The plant must be operated, serviced, and maintained by suitably trained and instructed personnel. Instructions and procedures given by the vessel operator to prevent and avoid potential dangers are compulsory.

Qualified personnel are by reason of their training, experience, instruction and their knowledge of the relevant standards, regulations, accident prevention rules and working conditions, authorized by the person responsible for the safety of the plant to perform the appropriate activities required, and thereby able to recognize and prevent potentially dangerous situations. Knowledge of first aid and local rescue organization arrangements must also be included.

Persons under the influence of drugs, alcohol or reaction-impairing medication must not be allowed to operate or work on the plant.

The personnel responsible for the installation, operation and maintenance of the equipment must be appropriately qualified. The operator must ensure that they have completely understood the instructions contained within this manual.

All warranties and guarantees with respect to the machine function and durability shall be void should the operator fail to adhere to these safety instructions or any other instructions within this manual.

2.6 Protective Equipment

Protective clothing such as safety helmet, safety glasses, safety gloves and safety shoes must be worn when applicable to working on the plant. Rubber gloves suitable for contact with wastewater must be worn.

2.7 Risk of Ignored Safety Instructions

There is a danger to life and limb for personnel if the plant is operated incorrectly or improperly.

Rotating parts and the energy released with a sudden stop can cause serious damage to personnel and equipment, which can also result in extended downtime and high repair costs with damaged parts.

The associated risks due to failure to adhere to these safety instructions include but are not limited to:

- Endangering people due to electrical, mechanical, and/or chemical/biochemical hazards.
- Endangering the environment due to leakage of hazardous material (where chemicals are involved).

Failure of important equipment and process functions leading to inferior performance.

2.8 Spare Parts

Only original spare parts and equipment authorised by FMD are suitable and safe for use.



Note: Incorrect or faulty spare parts can lead to damage, malfunction or complete breakdown of the equipment!

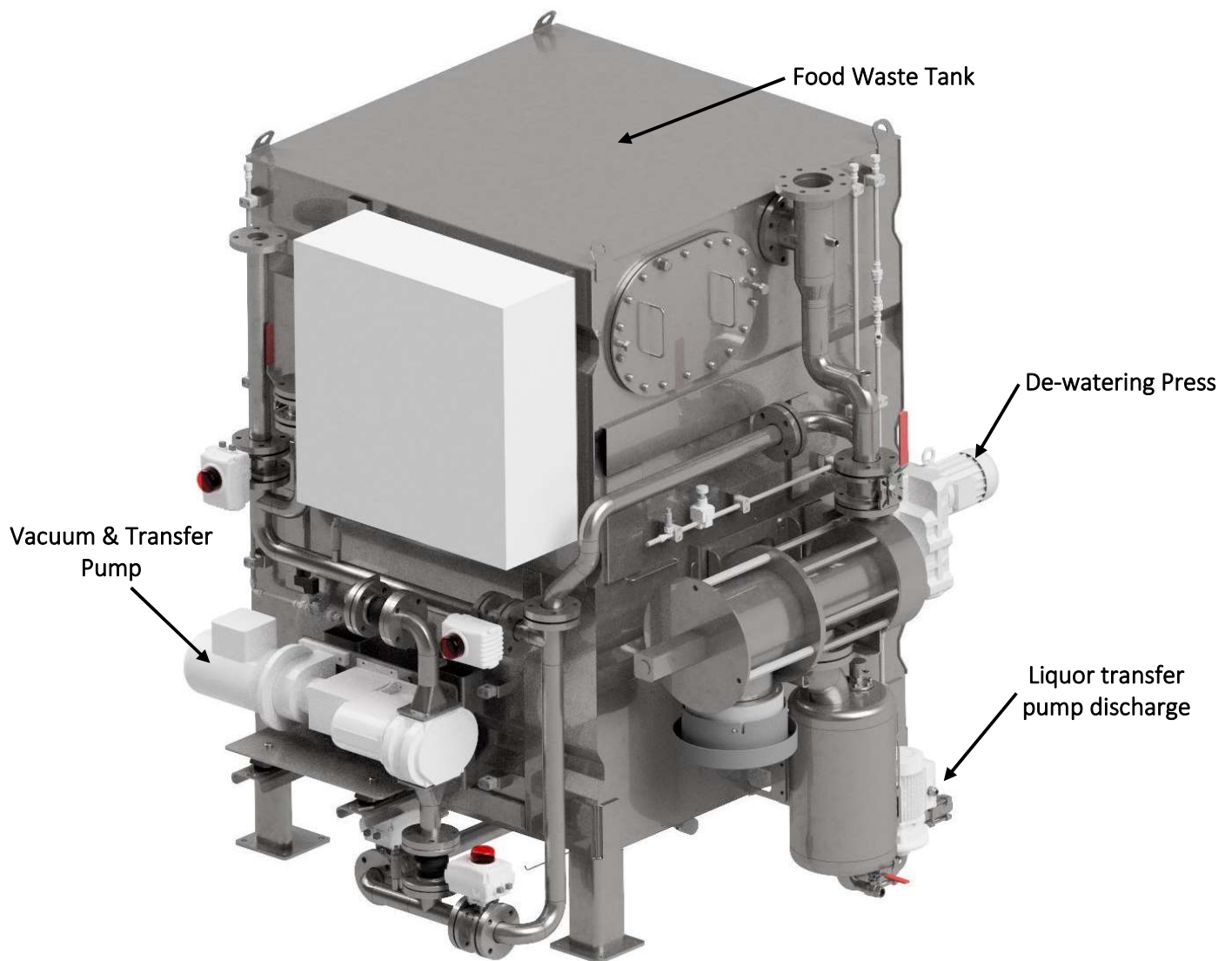
Installation of any parts or any modifications not authorised by FMD will nullify liability for any resulting damage or injury and change in performance, all guarantee, service, compensation, and liability claim against the manufacturer, its agents or representatives are forfeit.

3 System Introduction

3.1 Technical Data

Tank Capacity	1,200L
Pump Capacity	max 16,000L/hr
Dewatering Capacity	max 20,000L/hr
Voltage	400V 50Hz
Installed Power	~5 kW

3.2 Identifying Features of the Plant



FWC25 depicted

3.3 What it Does

The FWC system is designed to collect food waste from attached macerator stations using vacuum generated by the integrated pump and transfer the macerated slurry to the food waste dewatering press for separation of the solids from the liquid or to the holding tank for later processing or discharge overboard.

3.4 How it Works

When a macerator station lid is closed and the start button is pressed, the control system of the FWC12 opens the appropriate valves based on the selected mode and then starts the vacuum pump. When vacuum builds up in the piping system to the required level the macerator will start and the transport water solenoid will open, after a few seconds the macerator discharge valve will open and the macerated food waste will be drawn through the pipe work by vacuum until it reaches the pump.

Normal operation will see the food waste sent to the dewatering press immediately with the food waste tank remaining empty during normal operation, this prevents the anaerobic breakdown of any organic material sitting in the tank and the production of associated smells.

The food waste press separates the solids and the liquids at 150 microns, the liquid is transferred via pump to the ships grease trap. The solids fall by gravity into an endless bag that seals in all smells whilst the unit is in operation. The bag is monitored by a sensor that measures how full the bag is stopping the system and sending a common alarm when the bag requires changing.

Food waste can be discharged in accordance with the most recent revision Marpol Annex V regulations.

3.5 Don'ts

Do not attempt to process the following waste groups in the macerators or FWC12 system:

- Cooking oils & Fats
- Dough
- Cutlery, glass, crockery
- Plastic or solid waste
- Paints, aerosols, acids or alkali
- Chemicals
- Substances that can potentially lead to explosion or infection



Note: Equipment can be damaged if inappropriate items or materials are processed.

4 Installation

4.1 General

All safety regulations and instructions are to be observed.
The regulations and instructions in the Technical Manual must be observed.
Prior to commissioning FMD must be consulted.
It is recommended that commissioning be completed by a service technician from FMD.
The power supply may not vary from the contract specifications of the system.
The installation may not vary from the FMD approved installation drawings without prior written agreement.

4.2 Installation Drawings

The installation instructions should be used in association with the following drawings (where applicable) – See Section 8 Annex 5:

- Lifting and Transport Drawing
- Mounting and Pipe Connection Drawing
- Piping and Instrument Drawing (P&ID)
- Electrical Drawings

4.3 Transport

Lifting connections should only be made and executed as per the Lifting and Transport drawing supplied (where relevant). Transport and lifting should only be carried out with appropriate lifting equipment by trained persons. Attention is to be paid to all components extending past the main structure, so no damage occurs during lifting and transport operations.

4.4 Mounting

The FWC System is to be mounted in the upright position on the appropriate baseplates in accordance with the Mounting and Pipe Connection drawing supplied.



Note: FMD is to be consulted prior to any welding directly on the supplied plant.

4.5 Pipe Connections

The FWC System pipe connections are to be in accordance with the Mounting and Pipe Connection drawing supplied. FMD is to be consulted prior to modifications or changes.



Note: If the ship's piping is of CuNi the FWC system piping will have to be correctly isolated.

4.6 Electrical Connections

The FWC System electrical connections are to be in accordance with the Electrical Drawing supplied. FMD is to be consulted prior to modifications or changes.



WARNING: Electrical hazard! If not complied with, could lead to injury or death.

5 Commissioning

5.1 General

All safety regulations and instructions are to be observed.
The requirements and instructions in the Technical Manual must be observed.
Prior to commissioning FMD must be consulted.
It is recommended that commissioning be completed by a service technician from FMD.

5.2 Objective

The overall objective of commissioning is to ensure that the components of the system are complete, that installation is fit for purpose and the system is safe and ready to be set to work.

Commissioning requirements should include:

- Installation check of all main and auxiliary components.
- Check all interconnecting pipework is complete and correct with no leaks.
- Testing of all individual components and instrumentation for correct operation
- Testing of the control system to ensure correct function as per design and contract requirements.
- Establish plants operation in automatic and manual modes as required by contract documents and specifications.
- Confirm that all relevant acceptance test criteria have been met.
- Confirm that all alarms work as intended.
- Running of the system in auto for process establishment and evaluation.

5.3 Communication

Commissioning meetings with relevant stakeholders should be held as required. Topics should include:

- Commissioning program / schedules / procedures and checklists
- Commissioning resource requirements and allocation
- Commissioning progress to date
- Interfaces and interruptions with operating plant or external systems where they occur
- Interference issues between Construction, Commissioning and Handover
- Permit to Work requirements, operational requirements
- Planned major events, e.g. shutdowns, performance test periods
- Notice to relevant authorities
- Safety requirements

5.4 Supporting Documentation

A Commissioning Plan should be established referencing the following documents as required:

- Plant drawings (GA, P&ID)
- Installation, Operation and Maintenance Manuals
- Spare Parts List
- Materials Safety Data Sheets

5.5 Commissioning Personnel

- Commissioning Engineer from FMD
- Commissioning Engineer from Shipyard
- Ships' Engineer (Crew)
- Client Representative
- Project Manger

5.6 Pre-Commissioning

All equipment is on site installed as per relevant instructions, GA, schematics & drawings with electrical connections and tests completed.

Pre-commissioning starts after completion of the installation of the Plant and includes all activities that are needed to verify the correct installation of each individual component.

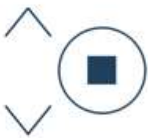
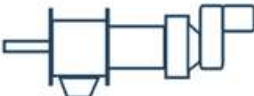
At the start of this phase the power should be connected to the main isolator and locked out by the commissioning engineer in charge.

- Check all supplies are connected correctly and isolated
- Undertake pre-commissioning checklists
- Close out any punch list items

6 Operation & General Maintenance

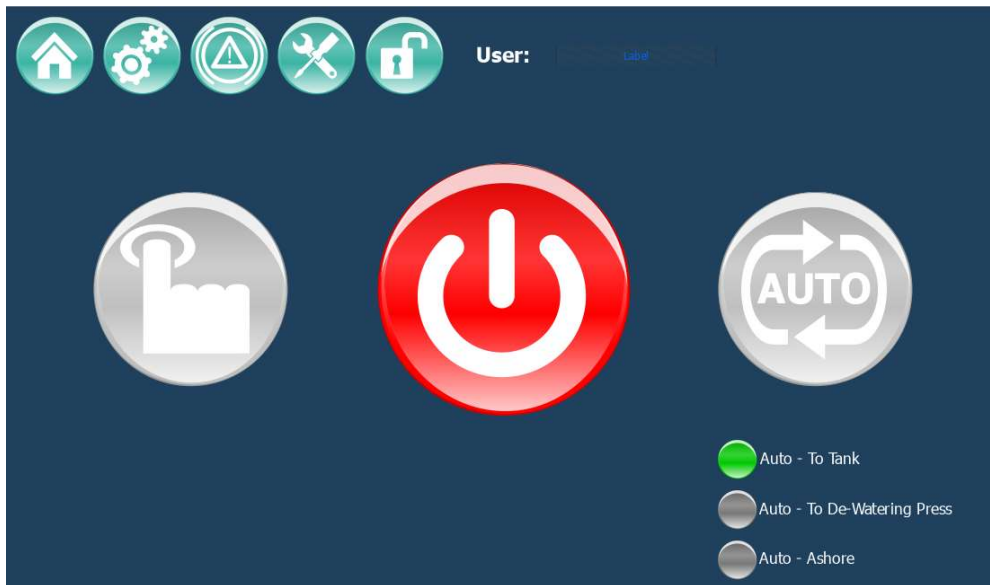
6.1 Navigation of the HMI

6.1.1 HMI Symbols

	Home		Page Forward/Back
	Settings		Manual
	Alarms		Stop All
	Maintenance		Automatic Run
	User Login		Manual Function Button
	Macerator		Valve
	Reversible Pump		De-Watering Press

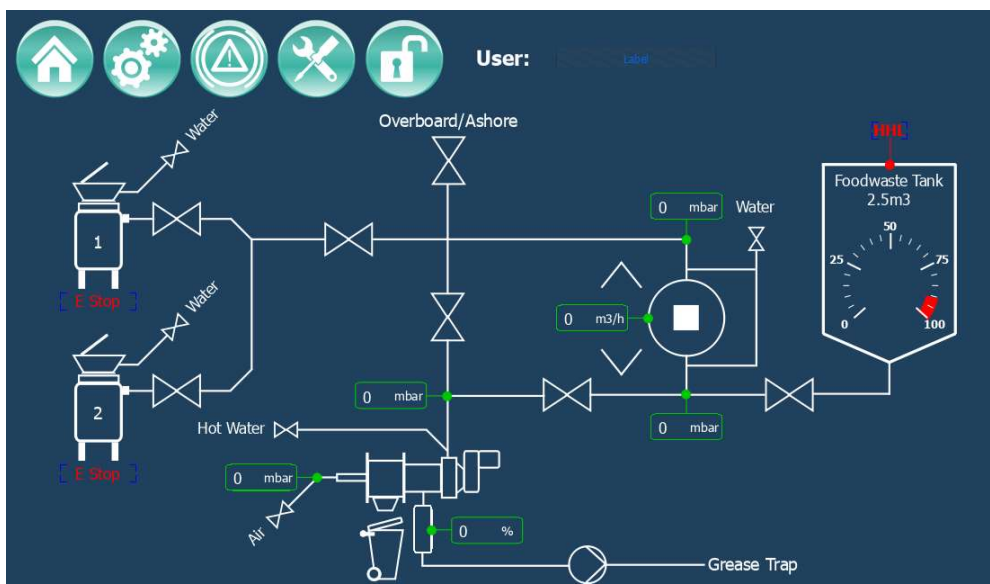
6.1.2 Home Page

Operation, control, monitoring, and parameter changes for the FWC are carried out from the touch panel (HMI) located on the front of the electrical cabinet.



6.1.3 Automatic Operation Page

Normal use mode with all control and safety systems active. From the home page select the *Auto* button on the screen, the Automatic Operation Page will appear on screen. Pressing the *Home* button will return to the home page where the red *Stop All* button is available to stop the plant running.

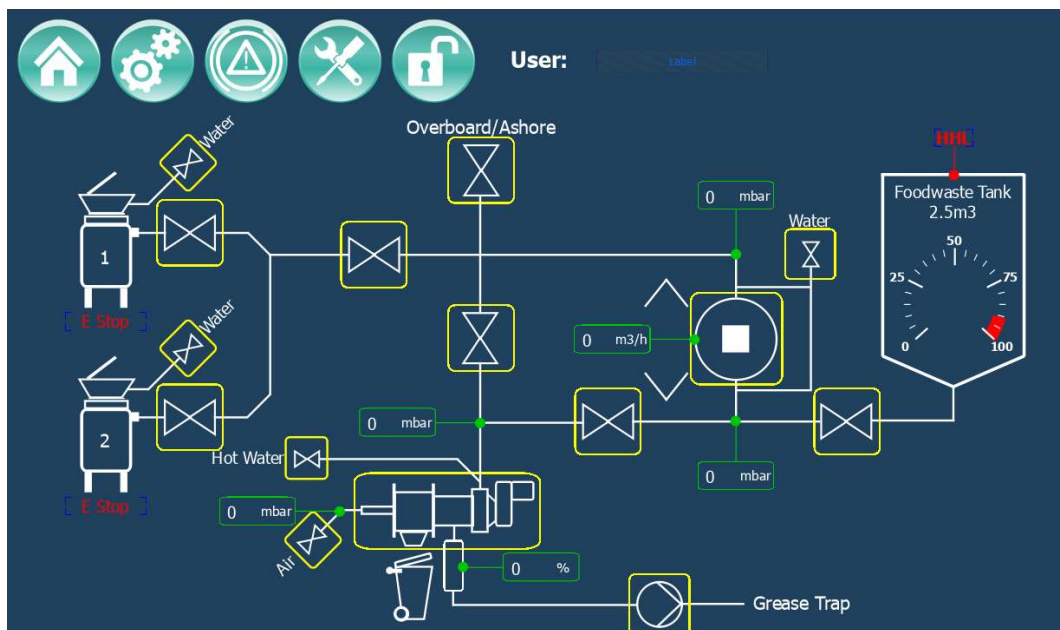


6.1.4 Manual Operation Page

All main components fitted to the system can be operated manually from the HMI. All components usable in manual appear on the screen with a yellow box highlighting them, this forms the *Start / Stop* button for the specific component.



CAUTION: When operating in manual it may be possible to start pumps with valves closed, particular care must be taken not to damage the plant with this type of operation.



6.1.5 Settings Page 1

Settings Page 1 displays a configuration interface for various sensors. The top navigation bar includes icons for Home, Settings, Alarm, Maintenance, and Lock, along with a User field and a back arrow. The main content area is titled 'Sensor Set Points' and contains a table with columns for LLL, LL, HL, and HHL. The table lists seven sensors with their respective set points in mbar or mm. A 'Sensor Time Delay' field is set to 0 seconds.

		LLL	LL	HL	HHL	
M.00.02.01	Level / Foodwaste Tank	000	000	000	000	mbar
M.00.03.01	Vac & Press / Foodwaste Pump	000	000	000	000	mbar
M.00.04.01	Vac & Press / Foodwaste Pump	000	000	000	000	mbar
M.00.05.01	Press / Dewatering Press Inlet	000	000	000	000	mbar
M.00.06.01	Press / Press Air Pressure Loss	000	000	000	000	mbar
M.00.07.01	Level / Bag Full Sensor	000	000	000	000	mm
M.00.08.01	Level / Foodwaste Liquor Tank	000	000	000	000	mbar

Sensor Time Delay: 0 secs

6.1.6 Settings Page 2

Settings Page 2 displays configuration options for pumps and macerators. The top navigation bar is identical to Page 1. The main content area includes fields for (Vacuum) Pump Rate, (Discharge) Pump Rate, (Dewatering Press Feed) Pump Rate, Dewatering Press Stop Delay, and Macerator Start Delay, all in m3/hr or sec's. On the right, there are three macerator status indicators, each with a red 'X' and a blue checkmark.

(Vacuum) Pump Rate: 008 m3/hr

(Discharge) Pump Rate: 008 m3/hr

(Dewatering Press Feed) Pump Rate: 005 m3/hr

Dewatering Press Stop Delay: 000 sec's

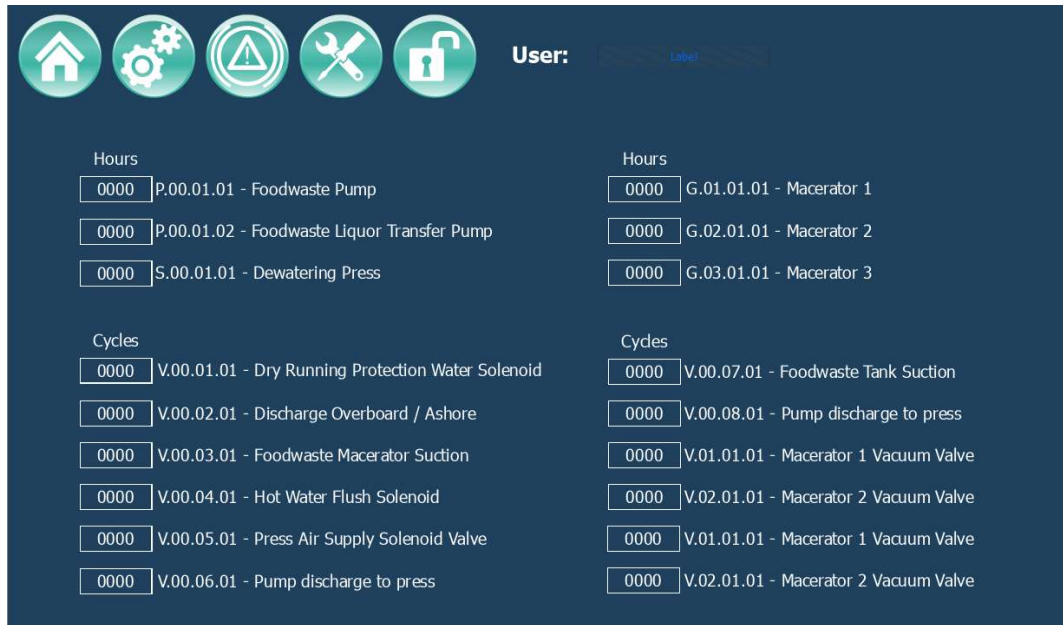
Macerator Start Delay: 000 sec's

Macerator 1: [Red X] [Blue Checkmark]

Macerator 2: [Red X] [Blue Checkmark]

Macerator 3: [Red X] [Blue Checkmark]

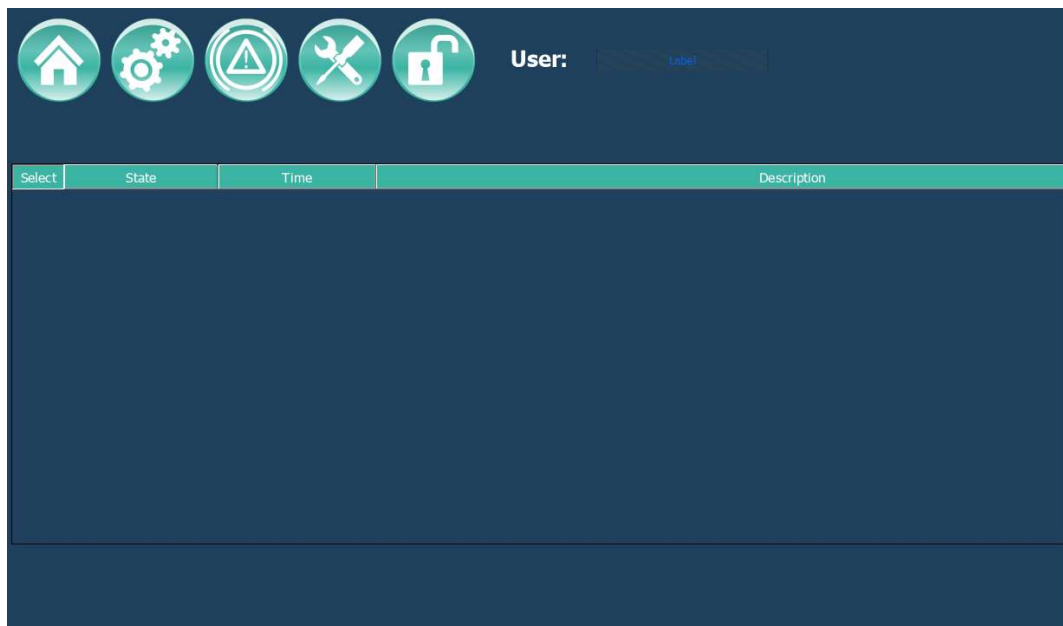
6.1.7 Maintenance Page



The screenshot shows a maintenance page with a dark blue background. At the top, there are five circular icons: a home icon, a gear icon, a warning triangle icon, a wrench and screwdriver icon, and a padlock icon. To the right of these icons is a 'User:' label followed by a text input field containing the word 'Label'. Below the icons, the page is divided into two columns. Each column has a 'Hours' section and a 'Cycles' section. Each section contains a list of items, each with a '0000' value in a text box and a description.

Hours	Cycles	Hours	Cycles
0000 P.00.01.01 - Foodwaste Pump	0000 V.00.01.01 - Dry Running Protection Water Solenoid	0000 G.01.01.01 - Macerator 1	0000 V.00.07.01 - Foodwaste Tank Suction
0000 P.00.01.02 - Foodwaste Liquor Transfer Pump	0000 V.00.02.01 - Discharge Overboard / Ashore	0000 G.02.01.01 - Macerator 2	0000 V.00.08.01 - Pump discharge to press
0000 S.00.01.01 - Dewatering Press	0000 V.00.03.01 - Foodwaste Macerator Suction	0000 G.03.01.01 - Macerator 3	0000 V.01.01.01 - Macerator 1 Vacuum Valve
	0000 V.00.04.01 - Hot Water Flush Solenoid		0000 V.02.01.01 - Macerator 2 Vacuum Valve
	0000 V.00.05.01 - Press Air Supply Solenoid Valve		0000 V.01.01.01 - Macerator 1 Vacuum Valve
	0000 V.00.06.01 - Pump discharge to press		0000 V.02.01.01 - Macerator 2 Vacuum Valve

6.1.8 Active Alarm Page



The screenshot shows an active alarm page with a dark blue background. At the top, there are five circular icons: a home icon, a gear icon, a warning triangle icon, a wrench and screwdriver icon, and a padlock icon. To the right of these icons is a 'User:' label followed by a text input field containing the word 'Label'. Below the icons, there is a table with four columns: 'Select', 'State', 'Time', and 'Description'. The table is currently empty.

Select	State	Time	Description
--------	-------	------	-------------

6.1.9 Alarm History

6.2 Operating Modes Food Waste Collection System

The day-to-day operator involvement required during normal operation is minimal and consists of basic daily observations, minor maintenance that has come due and response to any alarms.

Any changes or adjustments should be made by appropriately qualified persons and should ideally be recorded.

6.2.1 Auto to Tank

Select *Auto to Tank* on the home page then press the *Automatic Run* button, this will set the FWC12 plant to automatic operation whereby pressing of the macerator *Start* button will set the run sequence in operation and discharge the macerated food waste into the holding tank.

6.2.2 Auto to De-watering Press

Select *Auto to De-watering Press* on the home page then press the *Automatic Run* button, this will set the FWC12 plant to automatic operation whereby pressing of the macerator *Start* button will set the run sequence in operation and discharge the macerated food waste into the de-watering press.

6.2.3 Auto Ashore

Select *Auto Ashore* on the home page then press the *Automatic Run* button, this will set the FWC12 plant to automatic operation whereby pressing of the macerator *Start* button will set the run sequence in operation and discharge the macerated food waste to the bunker station.

6.2.4 Manual Operation



CAUTION: When operating in manual it may be possible to start pumps with valves closed, particular care must be taken not to damage the plant with this type of operation.

6.3 Operation Macerator



To start the macerator, it must be in the ready status, the *E-Stop* should not be illuminated, and the *Start / Run* button should be illuminated solid green. Once the macerator is filled with food and the lid is closed the press *Start / Run* button it will change from solid green to flashing green to show the macerator is active. It will follow the automated sequence programmed through to completion and the *Start / Run* button will return to solid green showing the macerator is complete and ready for the next batch.

7 Components

7.1 Macerators

7.1.1 Safety Precautions

Owner / User Responsibility

The owner and/or user must have a sound understanding of the operating instructions and warnings before using this equipment.

There are several forewarnings indicated throughout this manual that may stress a possible unsafe condition or important information regarding the use of this equipment.

The system operator must ensure that all relevant accident prevention regulations of their industry are observed and complied with. These safety regulations are only supplementary to the existing safety regulations of the entire system.

This manual has been prepared to provide you with a simple and understandable guide for equipment operation and maintenance. To keep your unit in top running condition, follow the specific maintenance procedures given in this manual.

General Warnings:



WARNING: Ensure you have read the safety definitions and symbols and understand all instructions before installing, operating, or servicing this machine. Failure to do so could result in severe personal injury or property damage.



CAUTION: When installing or maintaining the pre-screen, shut off and lock out power before removing any covers. Never make adjustments while in operation, except for those prescribed in this manual.



DANGER: Keep doors and guards in place at all times to avoid exposure to rotating parts. While the power is on, severe injury can occur if hands or feet reach inside the unit.



WARNING: A qualified electrician must perform all wiring. Failure to comply may result in severe personal injury or property damage.

Electrical System Precautions

The electrical equipment must be secured against unauthorised access and may only be tested and maintained by trained specialists.



WARNING: Electrical Hazard! ALWAYS check for no voltage before starting work!

When working on the switchgear and motors, the system must de-energized, tagged out and secured against re-connection in accordance with the ship's working practices and permit to work policy.

Biohazard

The food waste press can contain a variety of bacteria and viruses. In general, these are non-pathogenic strains, however, the inhalation of aerosols and contact with the skin and mucous membranes should be avoided.



WARNING: Biohazard! When working on the system the operator must wear suitable clothes, eye protection and rubber gloves suitable for contact with wastewater. Failure to do so could lead to illness or death.

The nature, control and extent of the necessary vaccinations and pre/post treatment examinations should be determined by the ship or company medical professionals.

7.1.2 Macerator Description

Normal Capacity:	850kg/hr
Installed Power:	4kW
Rated Amps:	9A (400V/3ph/50hz) 7.5A (440V/3ph/60hz)
Weight:	60kg



Discharge cone is pre-mounted with the following:

- water flushing nozzles
- solenoid valve G $\frac{1}{2}$ " 24Vdc
- inline strainer G $\frac{1}{2}$ ", with R $\frac{1}{2}$ " external thread
- safety interlock switch

7.1.3 What it Does

Food waste is fed into the machine and ground into small particles for transport by vacuum to the holding tank or dewatering press.

7.1.4 How it Works

Food waste disposers use flails mounted on a spinning plate taking advantage of centrifugal power to force food waste particles against a grinding ring breaking the food down into small particles.

7.1.5 Transport of the Macerator

If the machine is to be stored before installation it should be stored in a dry and clean room where the relative humidity may not exceed 60%.

Recommended storage temperature of -20 °C to + 60 °C.

The machine should not be lifted or moved by hand. During installation, we recommend using a pallet jack or forklift.

7.1.6 Mounting

In order to weld a strong joint and to be able to grind to an even and fine surface between the discharge cone and the working bench/tabletop, the cone is welded with its upper flange against the underside of the working top. The opening in the worktop must be at least 398 ± 2 mm in diameter. The disposer is then screwed into the lower attachment ring of the cone.

Mounting of models 520, 530, 550 and 575, delivered with legs as standard

The weights of models 520, 530, 550 and 575 require legs for support. At installation site the separately delivered 3 legs must be fixed to the 3 clamps in the disposer bottom plate.



When mounting the legs:

- Measure the height (X mm) between the flange of the mounting assembly and the floor/floor plate.
- Measure the height (Y mm) of the disposer without legs including the 3 mm rubber seal.
- Lay down the disposer carefully on its side. Slide the three legs into respective clamp. Each leg should protrude $X - Y - 4$ mm.
- Tighten the clamps. Make sure that the legs are fixated, and that their position have not changed during the tightening of the screw.
- Place the disposer so that it stands on its legs. Move the disposer carefully into position under the flange of the mounting assembly.
- Using the six screws with nuts fixate the disposer onto the flange with the 3 mm rubber seal in between. Tighten the screws alternately and evenly until the rubber seal is compressed approximately 1 mm.
- Use a jack or a lever to push up the disposer towards the flange of the mounting assembly.

While this upward pressure is maintained, untighten each leg and pull it downwards. Make sure that all legs are firmly pressed against the floor. Tighten the clamps using a torque of 17 Nm. As this is a critical matter, a torque wrench should be used. If no torque wrench is available use a lever of approximately 30 cm and a force of 6 kg.

7.1.7 Electrical Installation

The electrical connection must be done by an authorized electrician and in accordance to valid local regulations.

The wiring made at the factory, and the electrical connections to be done during installation, are shown in the electrical wiring diagram attached to this manual. The machine's specific electrical data are given in this manual, as well as on its serial number plate located on the outer cover of the macerator.

Supply Voltage

Check that the supply voltage to be connected corresponds to the specified voltage on the machine's serial number plate.

Check that the supply voltage for the delivered machine is fused and protected as specified in this manual.

Cables

Use connection cable having 1.5 mm² wires for machines having a rated current up to 14A. For machines having a rated current above 14A, use 2.5 mm² wires. The rated voltage and current is specified on the machine's serial number plate.

Direction of Rotation

The disposer operates correctly irrespective of the motor's rotational direction. It is therefore irrelevant in which sequence the electrical phases are connected to the disposer motor.

Safety Interlock Switch

Check the function of the interlock switch, if necessary, adjust the position of this switch and ensure it functions correctly.

7.1.8 Commissioning & Shutdown

Check before Start Up

Disconnect the power at the main isolator and lock out according to vessel lockout procedures. Check that the rotary shredder in the inlet opening of the disposer turns freely 360° in both directions by hand, and make sure that no foreign object has been dropped into the grinding unit during installation.

Checks during Start Up

- Start the disposer and determine that the grinder rotate.
- Check for automatic flushing water. The water should come directly when starting the machine.
- Check assembly, flushing water connections and drain connections for possible leaks.
- Make sure that the safety interlock switch functions and stops the machine if the lid is lifted.



DANGER: Keep doors and guards in place at all times to avoid exposure to rotating parts. While the power is on, severe injury can occur if hands or feet reach inside the unit.

7.1.9 Operation

Food waste that is difficult to grind, such as fibrous vegetables, tough fish skins and sinewy meat, should be broken down in size and mixed with other food waste.

Dry and sticky food waste (such as steamed rice, pasta, mashed potatoes, bread) should soaked in water before being fed into the machine.

Hard shells in various forms (e.g. from clams, oysters, crabs etc.) must not be fed into the machine as it among other things causes drainpipe blockages.

Also larger amounts of eggshells must not be fed into the machine as it also causes drainpipe blockages.

For batch fed machines of 2.5kW or more, and with tightly sealed top cover above the inlet for food waste, also any single larger sized bone can be mixed with other food waste when being fed into the machine.

Start and stop

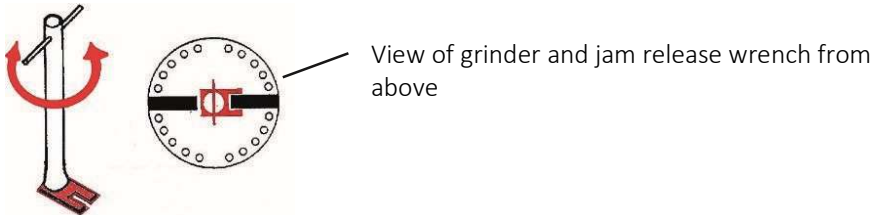
1. Open lid
2. Fill the feeding hopper to a maximum half of its volume
3. Close lid
4. Press the start button – disposer and flush water starts
5. When complete and the grinder has stopped wait a further 10 seconds before opening the lid to the feed hopper

7.1.10 Trouble Shooting

Disposer Reduces Speed, Stops or does not Start

A humming sound may be heard from the disposer motor.

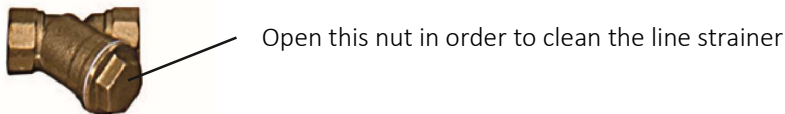
- Press the red stop button
- Use the main electrical isolator to disconnect electrical supply and lock it out
- Use protective gloves
- Open the disposer inlet lid
- Check if something is jammed in the disposer
- If something is jammed, place the jam release wrench on the center washer
- The recess on the lower plate of the wrench should grab one of the two bars on the disc of the grinding unit, as shown below



- Release the disc of the grinding unit by rotating the wrench backwards and forwards until this shredder disc rotates freely a complete turn in both directions. If needed, extend the bar on the wrench and hit the bar with a hammer/rubber mallet.
- Remove the jam release wrench.
- Remove any non-grindable object from the disposer grinding unit, e.g. cutlery.
- Close the disposer inlet lid
- Check and reset breakers & motor overload protection.
- Unlock and reset the main electrical isolator.
- Press start button and check the disposer functions correctly.

Disposer starts but there is no flushing water

- Is the water supply isolation valve open?
- Is a clicking sound heard when activating the water solenoid valve? If not, change the coil.
- Is the water strainer clogged? Isolate the water supply, open strainer and clean.



Machine does not Start and makes no Sound

- Check that the disposer inlet lid is in place and properly closed.
- Check that the main power isolator is in ON-position.
- Make sure breakers & motor overload protection are in order and switched on.

7.1.11 Maintenance

Maintenance Intervals

Description	Interval	Refers to
Cleaning of the machine	After daily use	
Check of the line strainer in the flush water pipe	First time after a month use, then when needed	
Preventive maintenance 1	First time after 1 month use, then after 1 year, and 3 yearly from then on	For this maintenance, please note that no parts need to be exchanged. Check electrical connections. Check pipe connections for leaks
Preventive maintenance 2	First time at 2 years, and 3 yearly from then on.	Preventive maintenance 1 Replacement of rubber seals with Position No. P14 and P17
Preventive maintenance 3	First time at 3 years, and 3 yearly from then on.	Preventive maintenance 1 & 2 Replace parts included in Service Package 1 shown in spare parts table. Inspect if the carrier (Pos. No. P2,) needs to be ordered and exchanged.
Replacement of parts due to wear	After approx. 9000 operating hours	Replace parts included in Service Package 2 shown in spare parts table.
Exchange of other parts	When needed	



WARNING: Ensure you have read the safety definitions and symbols and understand all instructions before installing, operating, or servicing this machine. Failure to do so could result in severe personal injury or property damage.



WARNING: Disconnect the power at the main isolator and lock out according to vessel lockout procedures.

Dismantling of Disposer

The below designation within brackets refers to the position numbers on the exploded view drawing and spare part list.

1. Isolate the power at the main isolator and lock out according to vessel lockout procedures.
2. Isolate the water supply to the disposer by closing a valve in the supply line.
3. Disconnect the disposer motor (**P1**) connection cable from the start & stop control unit.
4. Disconnect the disposer outlet.
5. Use a trolley with lifting plate or similar that can be inserted under the disposer. Detach the disposer from the assembly (6 screws and nuts, **P18**), and move the disposer to a suitable workbench for the continued dismantling.
6. Turn the grinder upside down and remove the housing (**P20**) by loosening the two screws (**P23**), and loosen the cable gland (**P22**) and insert the cable into the housing.
7. Mark on the end shield (**P3**) the position of the strap (**P19**) and then remove it by loosening two nuts (**P16**). Turn the grinder over again. Loosen the remaining two nuts with washers (**P16**).
8. These four nuts (**P16**) must always be changed for new ones before reassembly, as they are lock nuts. Note the location of the disposer outlet on the end shield (**P3**) relative to the hole pattern on the hood (**P15**) upper side, so that the outlet later may be reassembled in the same direction. For this purpose, is also an arrow engraved on the hood (**P15**) lower edge, which when reassembling shall be above the outlet of the the end shield (**P3**). Remove the hood (**P15**).
9. Remove the seal (**P14**) on the stationary shredder. This seal must always be replaced with a new seal when reassembled.
10. Pry the stationary shredder (**P13**) loose from the end shield (**P3**) using a crowbar that levers against the large tooth on the inside edge and is supported by the rotary shredder (**P10**) close to one of the two shredder blades. (See photo below.)



11. Loosen the axle screw (**P12**). If the rotating shredder (**P10**) keeps turning, and cannot be held still manually, use a pipe wrench to hold one of its shredder blades. Alternatively, lock the rotary shredder with a screwdriver placed through one of the holes at the disposer outlet. (See photo below.)



12. Remove the rotary shredder (**P10**) by placing two crowbars opposite one another under the edge of the shredder, supported by the side of the end shield (**P3**). (See photo below.)



13. Remove the V-ring seal (**P8**) and loosen the 4 fixing screws (**P7**) of the end shield (**P3**). Note the position of the outlet on the end shield (**P3**) relative to the terminal box for cable connection on the motor (**P1**), so that this outlet will receive the same direction when reassembling the disposer. Remove the end shield.
14. Inspect the washer (**P4**) for possible wear caused by the V-ring seal. Replace the washer if necessary.
15. Turn the end shield (**P3**) upside down and remove the locking ring (**P6**) by means of a circlip pliers. With some adequate protection in between (e.g. a piece of wood), apply pressure to the washer (**P4**) in order to get the two axle seals (**P5**) out. (See photo below.)



16. After this dismantling make sure that the surface of the sealing washer (**P4**) remains in good condition.

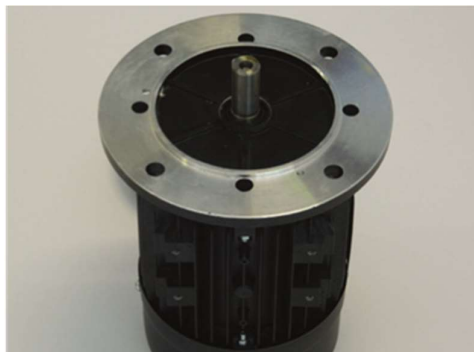
17. Examine the contact surfaces of the carrier **(P2)** for wear caused by the axle seals. There should only be two small symmetrical (barely noticeable) tracks on the carrier **(P2)** if it is to be reused. If the tracks are noticeable, asymmetrical or more than two, replace the carrier.



18. Terminate dismantling here if carrier **(P2)** and motor **(P1)** are not to be replaced. Start the reassembling of the disposer from **item 5 in section Assembly of Disposer** below.
19. **The V-ring seal (P8) and the two axle seals (P5) with special grease (P26) together with stationary seals (P14) and (P17) must be replaced at each dismantling.**
20. In Service Package 2 a disassembly screw **(P31)** for the carrier **(P2)** is included. Use this screw on top of the carrier and screw it down. This will remove the carrier from the motor axle. If needed, use the rotary shredder **(P10)** as a handle to prevent the motor axle from rotating. Alternatively, hold the carrier with a pair of pliers and a soft cloth in between.



21. When changing the motor **(P1)**, make sure that the new motor has the same classification and quality as the original. The motor must have two holes in the upper flange to allow for ventilation and drainage of water condensation (see arrows on photo below and original motor). When drilling these holes in the motor upper flange, it is important to drill at the correct angle, so that it does not penetrate the motor winding. All replacement motors delivered by the manufacturer have premade drainage holes.



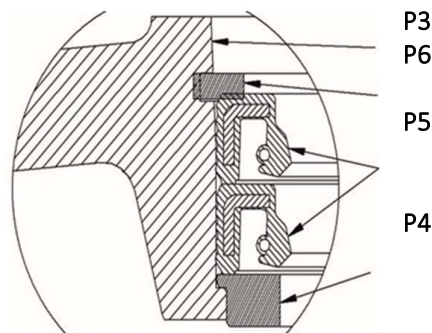
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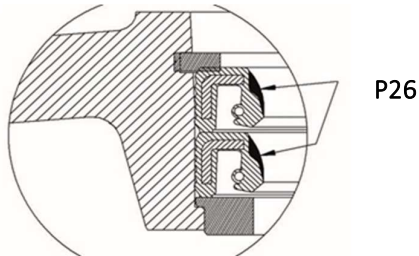
Assembly of Disposer

1. Clean all components carefully. Wipe the surface of the motor **(P1)** upper flange and its shaft. Make sure that there is no dust or grease residue.
2. Make sure that the motor shaft key **(P30)** is in the correct position. Apply a thin layer of manufacturer specified special grease **(P26)** on the motor shaft.
3. Wipe the carrier **(P2)** clean on the inside and outside.
4. Mark onto the motor shaft the depth of the hole in the carrier **(P2)**. From the top of the motor shaft, this measure is 38 mm for models 510, 515, and 48 mm for model 520, and 58 mm for models 530, 550 and 575. Carefully press the carrier **(P2)** onto the motor shaft all the way down to the mark. If hand power is not enough, use the axle screw **(P12)** as a mounting tool. If needed, use the rotary shredder **(P10)** as a handle to prevent the carrier from rotating.
5. On the end shield **(P3)**, clean the center hole for the axle seals **(P5)**.
6. Place the washer **(P4)** in the end shield with the flange facing towards the motor (see the sketch below under item 8, where the parts are seen with the end shield placed upside down as on the photo to the left).
7. Grease the sealing face of this center hole in the end-shield **(P3)** with special grease **(P26)**.
8. To avoid damaging the seals **(P5, 2pcs)**, press each seal one at the time into position in the end shield using a tool (for example a shaft end or other round bar, see below photo) that has the same outer diameter as these seals. The seals shall be placed with the opening and stainless spring upwards towards the washer **(P4)** according to the below sketch.

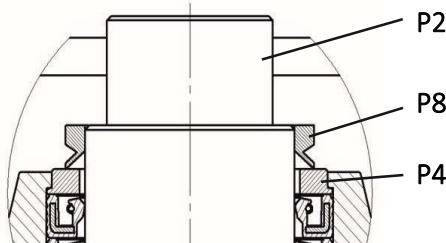


9. The seals are fixated by fitting the locking ring **(P6)** into the groove in the end shield **(P3)** according to the above sketch under item 8. Tap on this ring to ensure that it is seated properly in the groove in the end shield.

10. With a clean and soft putty knife fill the grooves along the sealing lips in the axle seals (**P5**) with special grease (**P26**). Be sure no dirt or particles happen to fall into this greasing.



11. With a clean soft cloth lubricate a thin layer of special grease (**P26**) on the carrier (**P2**) sealing surface for the axle seals (**P5**).
12. With reference to section 9.4, item 13 above, refit the end shield (**P3**) so that its outlet ends up 180° offset from the motor (**P1**) terminal box for cable connection. Take great care so that the lips on the axle seals (**P5**, 2pcs) not are damaged. For disposer models 510, 515 and 520 secure the end shield with tightening torque 47 Nm on the four screws (**P7**), and for models 530, 550 and 575 secure the end shield with tightening torque 81 Nm on the four screws (**P7**).
13. Grease the sealing lip of the V-ring seal (**P8**) and its seal washer (**P4**) using special grease (**P26**). Fit the V-ring seal over the carrier (**P2**) with its seal lip against the seal washer (**P4**).



14. Insert a small click of special grease (**P26**) into the keyway of the carrier (**P2**) and push on the key (**P9**).
15. Wipe clean the axle hole in the rotary shredder (**P10**). Also make sure that the mating surfaces on the end shield (**P3**) and the stationary shredder (**P13**) are clean.
16. Carefully fit the rotary shredder (**P10**) on the carrier (**P2**).
17. Wipe the carrier (**P2**) clean from excess grease.
18. Apply a layer of rubber sealing compound (**P27**) over the joint between rotary shredder and carrier (see photo below). Make sure that no sealing compound enters the screw hole of the carrier.



19. Fit the axle washer (**P11**) on the rotary shredder. Apply rubber sealing compound around the underside of the entire head of the axle screw (**P12**) and mount this screw. Make sure that sealing compound is squeezed out around the entire circumference of the axle washer, and also around the entire head of the axle screw. For disposer models 510 and 515 the axle screw is tightened with torque 9,3 Nm, and for model 520 the axle screw is tightened with torque 22 Nm, and for models 530, 550 and 575 the axle screw is tightened with torque 44 Nm.
20. Apply a thin uninterrupted string of rubber sealing compound (**P27**) in the seat of the end shield (**P3**) in which the stationary shredder (**P13**) should be pressed down. See below photo.



21. Carefully position the stationary shredder (**P13**) with its upper large tooth closest to the outlet of the end shield (**P3**).
22. Tap around the top of the stationary shredder (**P13**) with a plastic hammer to fix it in the end shield (**P3**), and make sure that the stationary shredder is all the way down in the seat of the end shield.
23. Secure the stationary shredder (**P13**) to the end shield (**P3**) by hammering 6 punch marks on the end shield with even distances around the joint between these two articles. See below photo.



24. Fit a new rubber seal (**P14**) on the top of the stationary shredder (**P13**). Make sure that it is centered.
25. Wipe clean the surface of the hood (**P15**) that faces the rubber seal (**P14**).

26. Carefully fit the hood (**P15**). For the upper hole pattern of the hood to end up in the same way as before dismantling, the outside arrow mark on the hood lower edge shall be positioned centrally above the waste outlet on the end shield (**P3**). This to make sure that the disposer outlet will have the same direction towards the sewer connection as before. Also feel with your fingers that the seal (**P14**) still is centered.

27. Secure the hood (**P15**) with 2 pcs. of washers and 2 pcs. of new unused locking nuts (**P16**) in the two positions that have no previously made markings for the strap (**P19**, see section **Dismantling of Disposer, item 7 above**). Tighten these two lock nuts (**P16**) only a little so that the seal (**P14**) is fixed. The final tightening is made after the strap (**P19**) has been fitted as per item 29 below.

For disposers installed in cabinet models (e.g. model 500-MB and 500-MC) the parts with position number **P19** up to and including **P25** are not used. For these disposers secure the hood (**P15**) with 4 pcs. of washers and 4 pcs. of new unused locking nuts (**P16**). Tighten all these lock nuts crosswise with torque 9,5 Nm for models 510, 515 and 520, and with 21 Nm for models 530, 550 and 575. Continue the assembly of the disposer installed in cabinets from item 34 below.

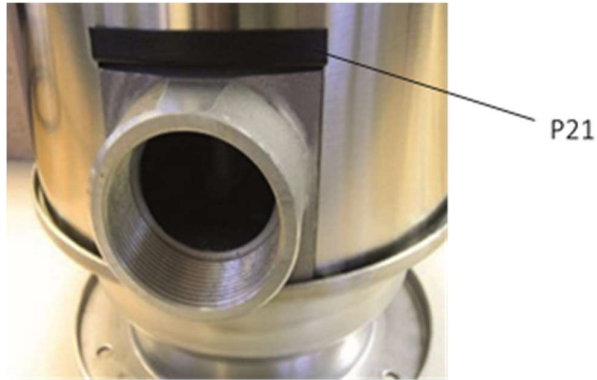
28. Turn the disposer upside down.

29. Place the strap (**P19**) diagonally over the two remaining screws and fix it with 2 pcs. of washers and 2 pcs. of new unused locking nuts (**P16**). Tighten all four lock nuts crosswise. For models 510, 515 and 520 with torque 9,5 Nm and for models 530, 550 and 575 with 21 Nm.



30. Pass the motor cable through the cable gland (**P22**) in the disposer housing (**P20**).

31. Mount the disposer housing (**P20**) over the motor (**P1**). While the housing is being lowered over the motor, pull the cable so that the remaining part of the cable comes out through the cable gland. Make sure that the sealing strip (**P21**) is in the right place on the housing. Also make sure that the six small “heels” of the housing (**P20**) end up inside the lower edge of the hood (**P15**).



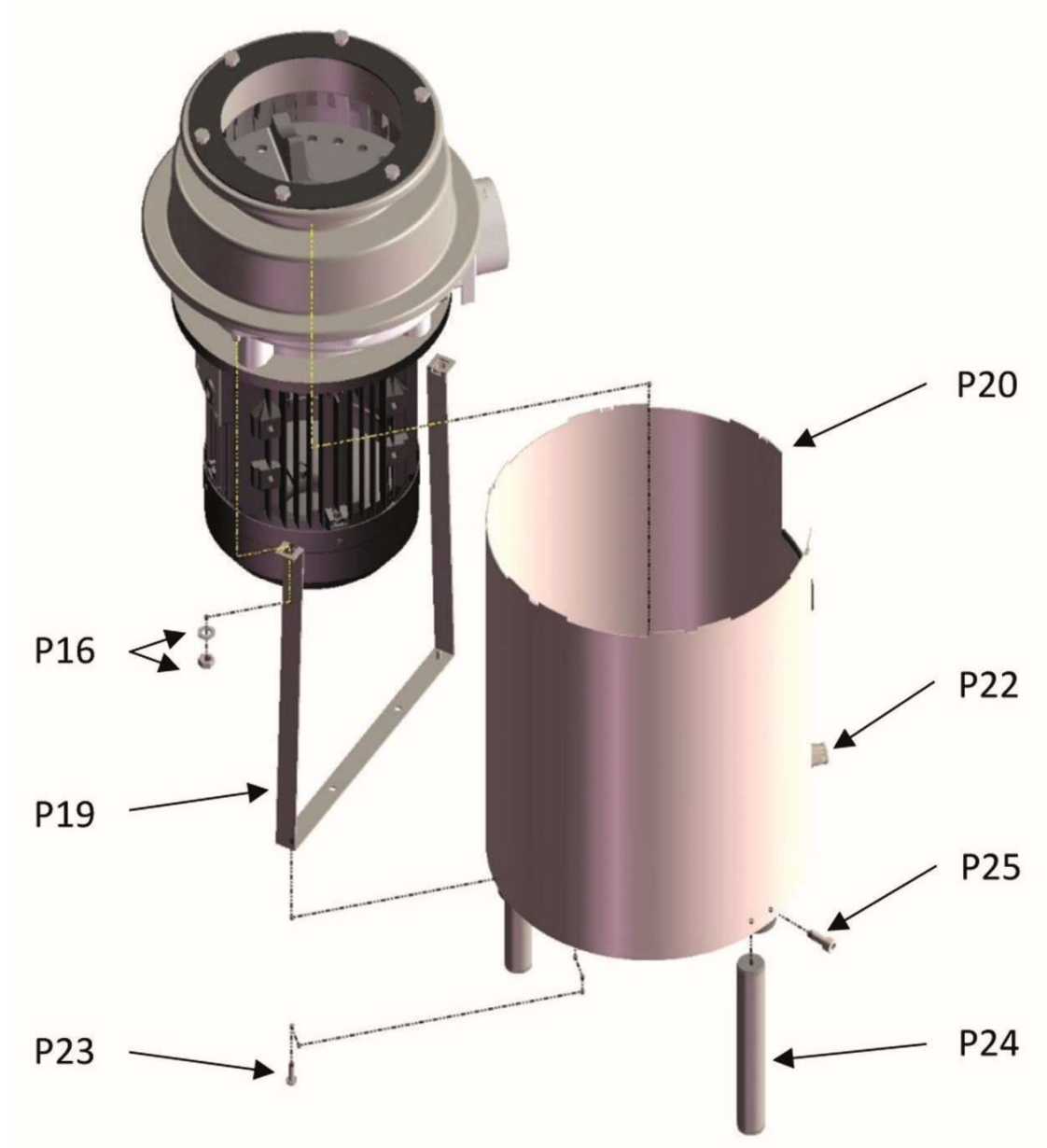
32. Fasten the housing (P20) to the strap (P19) with 2 pcs. of screws and 2 pcs. of washers (P23).

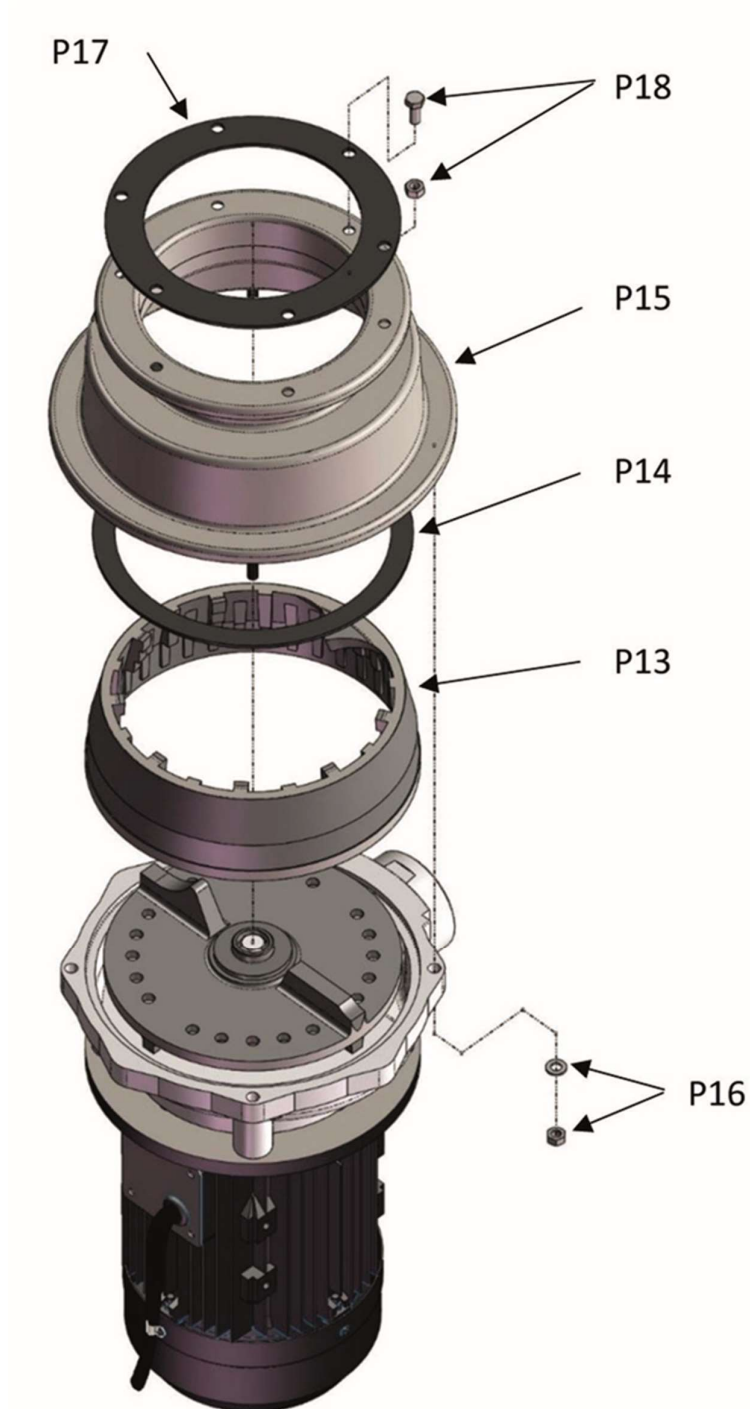


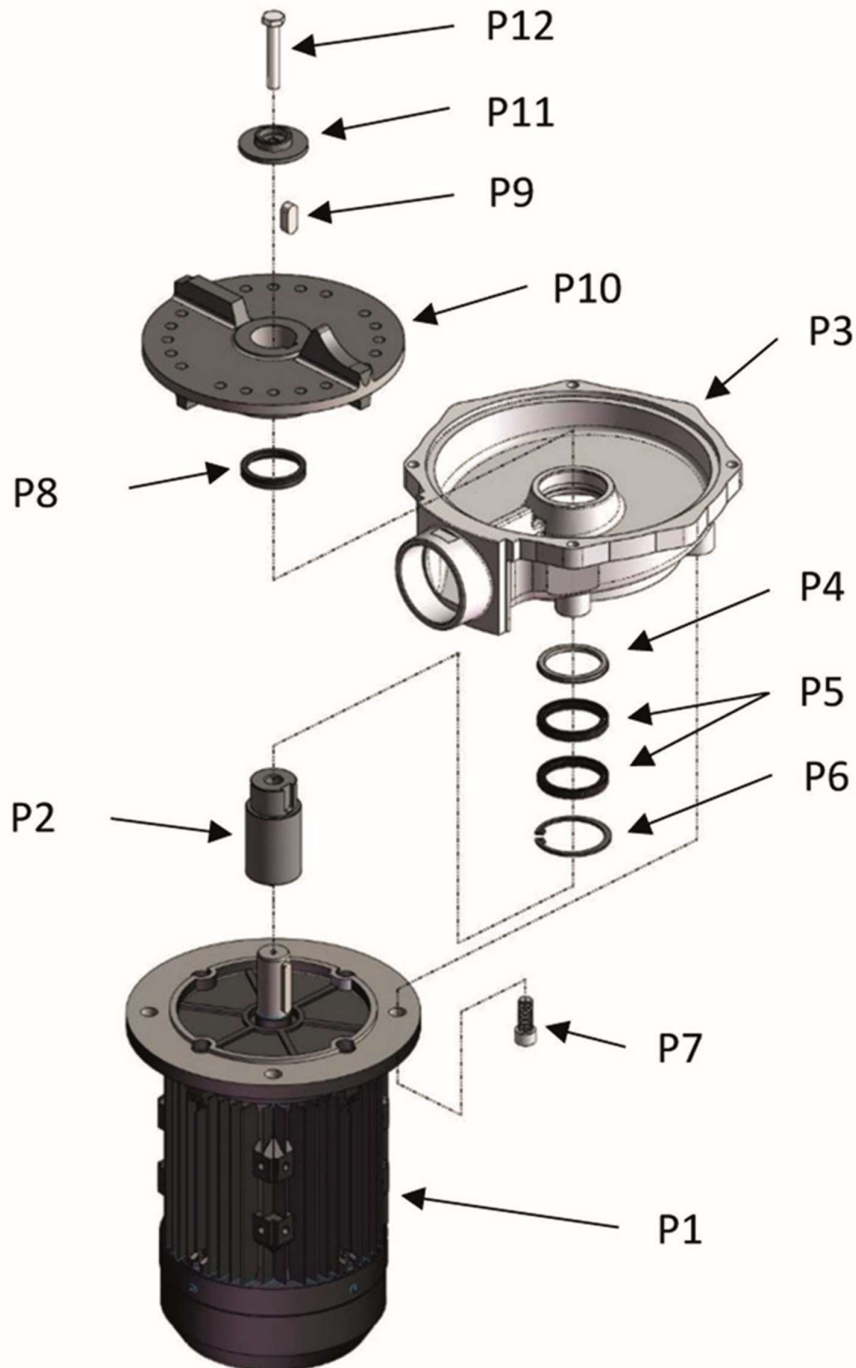
33. Pull out the cable as far as it goes through the cable gland. Then feed back about 5 cm to get a good relief of the cable. Lock the cable by tightening the nut of the cable gland (P22). Turn the disposer over again.
34. Check with a blade measure that the stationary seal (P14) is properly fixed between stationary shredder (P13) and hood (P15).
35. The disposer is now ready to be remounted under the assembly using the same trolley with lifting plate (or the same steady jack with lifting plate) as previously used when the disposer was dismantled. Be aware of the motor power supply cable so it does not get pinched and damaged.
36. Make sure that the seal for the hood (P17) is placed correctly before you move the disposer into position under the assembly for mounting with 6 pcs. of screws and nuts (P18).
37. Refit any disposer fixation to the wall/bulkhead or floor/floor plate.
38. Reconnect and seal the water trap with sewer pipe to the outlet on the disposer end shield (P3).
39. Reconnect the disposer motor (P1) connection cable to the start & stop control unit according to the electrical connection diagram supplied with the machine from Disperator.
40. Reconnect electrical supply using the main electrical safety breaker on the wall/bulkhead.
41. Reconnect the flushing water supply to the disposer by open the valve in the supply line.
42. Start up and test the disposer according to applicable start-up procedures described in section 6.5 of this user manual.

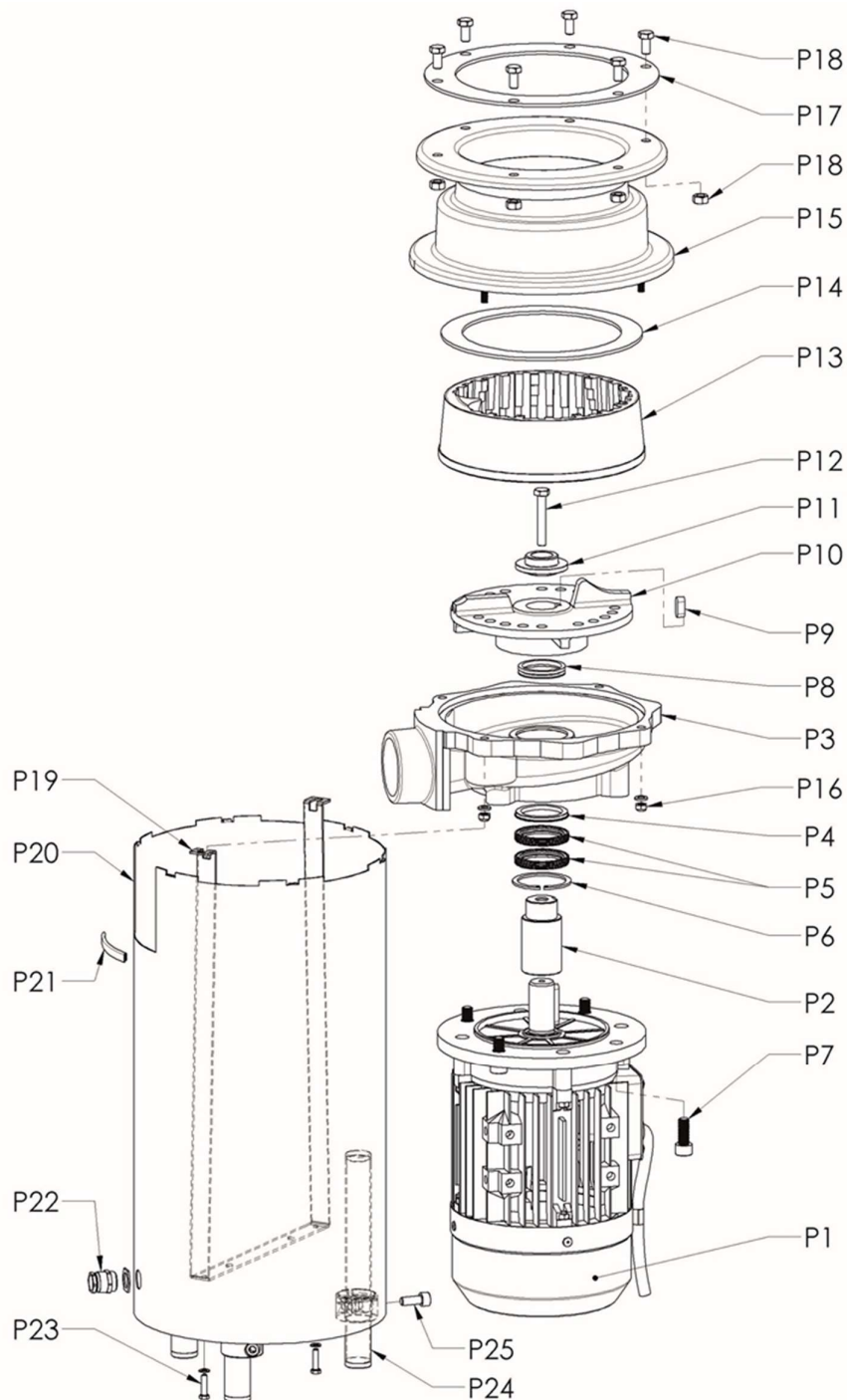
7.1.12 Spare Parts

Exploded Views and Spare Parts List for the Disposer









Position No:	Qty:	Denomination:	Spare Part No:	Included in Service Package:
P1	1	Motor with drained upper flange	-14	
P2	1	Carrier	-18	2
P3	1	End shield	-13	
P4	1	Washer for V-ring seal	-03/00	1 and 2
P5	2	Axle seal, with stainless spring	-03/02	1 and 2
P6	1	Locking ring	-03/03	1 and 2
P7	4	Screw for end shield assembly	-23	1 and 2
P8	1	V-ring seal	-03/01	1 and 2
P9	1	Key for carrier	-10	1 and 2
P10	1	Rotary shredder	-02	2
P11	1	Axle washer	-06	2
P12	1	Axle screw	-07	1 and 2
P13	1	Stationary shredder	-01	2
P14	1	Seal for stationary shredder	-05	1 and 2
P15	1	Hood	-16	
P16	4	Locking nut and washer for hood assembly	-25	1 and 2
P17	1	Seal for hood flange	-08	1 and 2
P18	6	Screw and nut for disposer assembly	-22	1 and 2
Below parts with Pos. No. P19 – P25 are not used for disposers installed in closed cabinet models				
P19	1	Strap for housing	-17	
P20	1	Housing	-15	
P21	1	Sealing list for housing	-15/01	
P22	1	Cable gland with nut	-19	
P23	2	Screw and washer for housing assembly	-26	1 and 2
P24	3	Adjustable disposer leg (only for Model 520, 530, 550 and 575)	-29	
P25	3	Screw for disposer leg	-29/01	1 and 2

Position No:	Qty:	Denomination:	Spare Part No:	Included in Service Package:
Not depicted parts				
P26	1	Special grease for seals and carrier	-04	1 and 2
P27	1	Rubber sealing compound	-09	1 and 2
P28	2	Axle seal, motor drive end and motor non-drive end	-14/01	
P29	2	Bearing, motor drive end and motor non-drive end	-14/02	
P30	1	Key for motor shaft	-14/03	
P31	1	Disassembly screw for carrier	-18/02	2
Parts depicted on the next page				
P17	1	Noise damping seal (alternative to spare part No. -08)	-08/1	
P32	1	Torque protection bar for fixation of disposer in wall/bulkhead	-28/V	
P33	1	Jam release wrench for rotary shredder	-31	
P34	1	Solenoid valve, G½" (BSPP), internal thread, incl. coil, cable plug and fixating washer	-35	
P35	1	Coil for solenoid valve	-36	
Not depicted parts				
P36	1	Line strainer, G½" (BSPP), internal thread	-37	
P37	1	Start & stop unit incl. contactor with motor overload protector	-38	
P38	1	Contactor with coil	-38/1	
P39	1	Start contact	-38/2	
P40	1	Motor overload protector	-38/3	

Always state machine serial number and spare part number (see above) when ordering spare parts.



Take Note: That the correct quality and strength of all nuts and screws used are important. FMD can not guarantee the safety if other screws and nuts than those ordered from us are used.

		
P17	P32	P33
		
P34	P35	P36

7.2 Food Waste Press

7.2.1 Safety Precautions

Owner / User Responsibility

The owner and/or user must have a sound understanding of the operating instructions and warnings before using this equipment.

There are several forewarnings indicated throughout this manual that may stress a possible unsafe condition or important information regarding the use of this equipment.

The system operator must ensure that all relevant accident prevention regulations of their industry are observed and complied with. These safety regulations are only supplementary to the existing safety regulations of the entire system.

This manual has been prepared to provide you with a simple and understandable guide for equipment operation and maintenance. To keep your unit in top running condition, follow the specific maintenance procedures given in this manual.

General Warnings:



WARNING: Ensure you have read the safety definitions and symbols and understand all instructions before installing, operating, or servicing this machine. Failure to do so could result in severe personal injury or property damage.



CAUTION: When installing or maintaining the pre-screen, shut off and lock out power before removing any covers. Never make adjustments while in operation, except for those prescribed in this manual.



DANGER: Keep doors and guards in place at all times to avoid exposure to rotating parts. While the power is on, severe injury can occur if hands or feet reach inside the unit.



WARNING: A qualified electrician must perform all wiring. Failure to comply may result in severe personal injury or property damage.

Electrical System Precautions

The electrical equipment must be secured against unauthorised access and may only be tested and maintained by trained specialists.



WARNING: Electrical Hazard! ALWAYS check for no voltage before starting work!

When working on the switchgear and motors, the system must de-energized, tagged out and secured against re-connection in accordance with the ship's working practices and permit to work policy.

Biohazard

The food waste press can contain a variety of bacteria and viruses. In general, these are non-pathogenic strains, however, the inhalation of aerosols and contact with the skin and mucous membranes should be avoided.



WARNING: Biohazard! When working on the system the operator must wear suitable clothes, eye protection and rubber gloves suitable for contact with wastewater. Failure to do so could lead to illness or death.

The nature, control and extent of the necessary vaccinations and pre/post treatment examinations should be determined by the ship or company medical professionals.

7.2.2 Food Waste Press Description

The FMD food waste press is exclusively designed for separating solids from wastewater at a maximum inlet flow rate of up to 20 m³/hr (the 'intended use'). Intended use also includes compliance with the operating, maintenance and servicing conditions prescribed in this document.

The press must not be used for flow rates higher than those specified and must not be subjected to inlet pressure in excess of 100mbar.

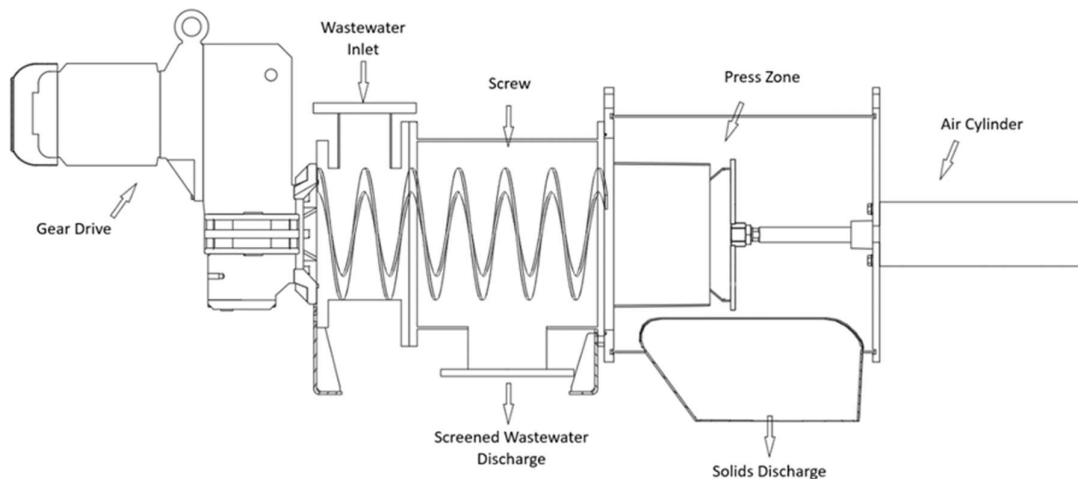
Any use above and beyond the "intended use" is considered improper and the manufacturer assumes no liability for damage resulting from this, such risk is borne solely by the operator.

Technical Data

Press Type	TSP20
Serial Number	221010004Z507
Drive Type	BF30
Drive Specification	400V / 50Hz
Year of Manufacture	2020
Equipment Weight	~150kg
Body/Door/Press Zone Material	1.4571 Stainless Steel
Shaft & Screw Material	1.4571 Stainless Steel
Filter Basket Material/Mesh Size	1.4571 Stainless Steel / 150 micron
Seal Material	EPDM

7.2.3 Identifying Features of the Food Waste Press

General Construction Section



7.2.4 What it Does

The press separates the solids from wastewater by means of a 150 micron screen and a rotating screw. The screened liquid falls by gravity to the intermediate tank and the solids fall into the wheelie bin, via the solids discharge chute, into an endless plastic bag.

7.2.5 How it Works

The press has been designed specifically for screening solids from wastewater and sludge streams. The inlet wastewater should be such that it freely flows into the press without being pressurized.

The principle is simple: a slow rotating screw is enclosed in a basket with a “press zone” integrated at the end which is controlled and adjusted automatically by the air cylinder load on the conical end cap.

The liquid passes through the openings of the screen basket while the slowly rotating screw transports the retained solids to the press zone.

In the press zone the material is compressed, de-watered and finally pushed through the conically tapered end cap to remove the solids with a consistent dryness quality.

The solid content is a function of the input material, the installed basket screen size, and the press zone cone pre-load (which is adjusted using a precision air pressure regulating valve).

The pressed solids drop by gravity through the solids discharge chute into the endless bag which is contained in the wheelie bin.

7.2.6 Installation of the Food Waste Press

Lifting & Positioning

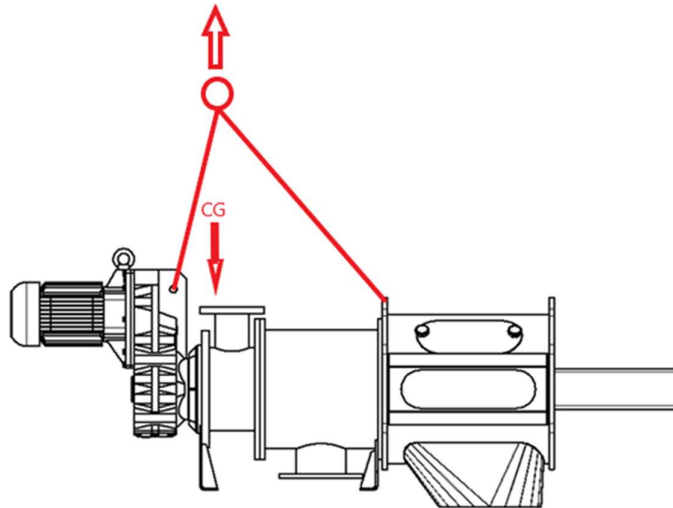


WARNING: During lifting and transporting of loads, there is the risk that lifting accessories may fail. Never step under a suspended load, keep a safe distance and cordon off the danger area.

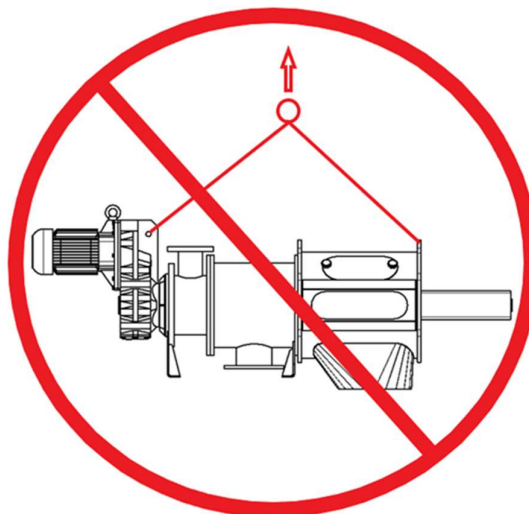


CAUTION: Ensure that the used lifting equipment is adequate for the load specified.

When lifting the press with a crane or chain block, it is only permitted to lift the machine by the provided holders:



Under no circumstances is the machine to be lifted by the pneumatic cylinder or the front plate eye:



7.2.6.1 Mounting

The frame to which the press is bolted must be flat and horizontal. If this is not the case, the screw separator alignment will be distorted when bolted down to an uneven surface – and this will potentially damage the screw separator and invalidate the warranty.



CAUTION: *If the press is solidly bolted to an uneven surface there may be severe misalignment that can permanently damage the press! Inform FMD immediately and carefully document the incidence.*

7.2.6.2 Electrical Installation

If all mechanical connections are examined and satisfactory, the drive can be connected according to the switching plan attached to the terminal box.

Thermal motor protection (T1 and T2) is only optionally installed.

The press must be earthed according to the ship's electrical requirements.



WARNING: *For the electrical connections, the relevant safety regulations should be followed! Electrical installation must be in accordance with relevant standards and completed by a qualified electrician.*



CAUTION: *The correct direction of rotation is to be observed; this is clearly marked by the red arrow on the drive. Incorrect direction of rotation can lead to the damage of system parts!*



Picture: Direction of Rotation

7.2.7 Commissioning & Shutdown

7.2.7.1 Pre-commissioning Checks

Prior to the first test run, all points below should be verified:

- Are all safety provisions available in accordance with the regulations?
- Are all electrical connections made, tested and accepted according to the standards?
- Are all transport locks removed?
- Are the inlet and outlet pipes connected?
- Open the inspection covers of the screen and check for any foreign objects such as construction or mounting materials, tools, rags etc.

7.2.7.2 Initial Test Run



WARNING: Risk of crushing / rupture of limbs! Ensure the inspection covers are fitted and secured before starting the machine.

If all the pre-commissioning points are checked and found in order, then the test run can be started by following the below procedure:

- The press gearmotor should be bumped without water to verify the direction of rotation is correct. The press must not be operated in the wrong direction for more than a few seconds.
- Before the first test run with water, the system must be ensured clear of debris, tools, sharp-edged objects, iron filings or other objects located upstream of the inlet. Such objects can damage the press.
- For the first test run with water, the air pressure regulating valve for the press zone should be set to ~1.5 bar. Check that the cone seals the press zone, a drip every 5-10 seconds is normal at this stage.

When starting up the press, liquid may leak briefly from around the press zone gasket.

For safety purposes protect yourself and other from any liquid that may potentially spray out if the doors on the discharge chute are open.

For the first run using wastewater do not run the feed pump at full power. Leave the TSP running for 30-60 minutes with the wastewater inlet flow at ~30% of maximum capacity during this period:

- Check the press for leaks.
- Check the solids bag is not filling with water.
- Observe the press discharge cone for leakage initially and later when using wastewater for movement to indicate a solids plug is forming.
- If the separation and the pressing result are within the required range, the inlet flow rate can be increased slowly.

7.2.7.3 Setting & Optimising the Press Discharge

As general rules:

- Never set the air pressure higher than 2.0 bar.
- Once the plug is established the optimum air pressure is generally 0.6 – 0.8 bar for GW/BW and 1.0-1.5 bar for food waste.
- The higher the air pressure the higher the chance a blockage may form in the press, which in turn leads to increased wear, break down and possible irreparable damage.
- Observe the press for a while. Reduce or increase the pressing force until the solid discharge runs evenly and continuously.
- If the press begins to vibrate, this is often a sign of excessive pressing force and an impending blockage.

7.2.7.4 Shutdown

- Turn off and isolate the inlet supply pumps to prevent them starting.
- Wait until the feed pipes have emptied.
- Reduce the pressing force and move the compressed air cylinder to the rear.
- Let the separator run for a short while longer until the solids in the machine have been completely discharged.
- Turn off the press motor and protect against inadvertent re-starting.
- The system must be thoroughly cleaned.
- Put up signage to notify that the press has been taken out of operation.



Take Note: If the press is to be idle or shut down for more than 72 hours, there is a risk that the solids plug (formed against the cone) will begin dry out and solidify – see below:



If the plug has dried out and the press is re-started, after a short time of operation the pressure behind the plug will increase until it releases suddenly - and there is the potential for wastewater to carry over into the solids bag.

Avoid unexpected stress testing of the solids bag.

Open the service port, retract the cone and remove the solidified solids plug before re-starting the press.

Picture: Solids bag filled with wastewater

7.2.8 Maintenance

Overview & Maintenance Intervals

To maintain operational readiness, possible damage should be detected at an early stage. To preserve warranty and guarantee entitlements the operator is obliged to carry out regular inspection and maintenance work on the system in accordance with the respective operating and maintenance instructions as detailed in 7.1.10 Failure to observe these regulations will lead to the invalidation of the warranty. Operating persons must be informed of, and understand, all safety requirements before beginning start-up and maintenance work. For all maintenance, inspection and repair works the press is to be shut down and isolated with the main power isolator locked out in accordance with the onboard regulations and permit system.



CAUTION: Pay attention to all safety instructions during all maintenance and servicing work and the safety policies of the vessel.



WARNING: Before working on the press, isolate the power supply and lock out or remove fuses. There is a risk of crushed hands and limbs from the rotating shaft/screw in the drive area.

All system parts and operating media connected upstream of the press must be protected against unintended start.

7.2.9 Modifications to the Press

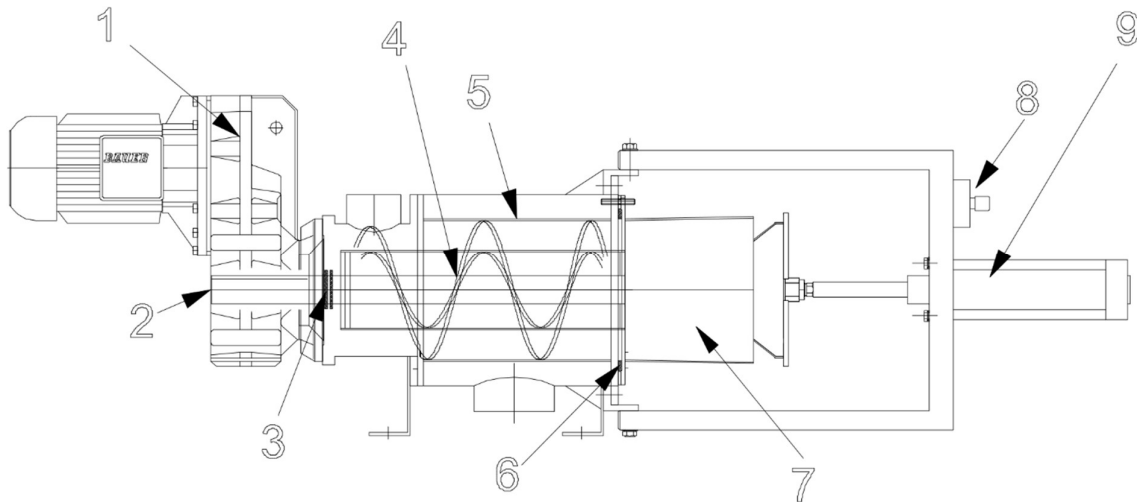
No modifications, attachments or rebuilding of the press may occur without the prior written authorisation of FMD.

Machine parts that are not in a safe usable condition are to be replaced immediately.



Take Note: Only original spare and wear parts may be used. Other parts are not warranted.

7.2.10 Spare Parts



Pos Nr.	Part Description	Part Nr.
1	0.75 kW Drive, Type BF30, 400V-50Hz	A00168
2	Main Shaft	A00169
3	Main Shaft seal, 2 per main shaft	A00170
4	Screw, Carbide plated	A00171
5	150 Micron 316L SS Press Basket	A00133
6	Tumbling Ring	Consult FMD
7	Pressing Zone	Consult FMD
8	Precision Venting Regulator	A00080
9	Air Cylinder for Press Zone	A00173
10	Precision Venting Regulator repair kit	A00332

7.2.11 Preventive Maintenance

The instructions for all visual inspections, maintenance and repair work must be observed.



WARNING: Before working on the TSP, isolate the power supply and lock out or remove fuses. There is a risk of crushed hands and limbs from the rotating shaft/screw in the drive area.

The essential maintenance for the TSP is summarised in the Preventive Maintenance Tasks table where the following abbreviations/codes are used:

- D** Daily maintenance work
- W** Weekly maintenance work: performed as required, latest after 100 operating hours
- M** Monthly maintenance work: performed as required, latest after 450 operating hours
- Q** Quarterly maintenance work
- S** Half-yearly maintenance work
- A** Annual maintenance work: performed as required, latest after 5000 operating hours

D	W	M	Q	S	A	Task	Reference
General Maintenance Work on the Press							
X						General visual inspection daily or after period of particularly high load	
	X					Check basket for fat, fibre growth or blockages & clogging	
		X				Check the mechanics for possible malfunctions of the drive, shaft bearing, cone etc	
		X				Check all nuts, screws, bolts and connecting elements for tightness	
					X	The complete mechanics, connections and bearings must be carefully checked and rectified as required.	
Maintenance work on sieve, screw							
	X					Check that the screw runs evenly in the basket	
Screw drive							
		X				Check gearbox for leaks	See gearbox Annex
				X		Change the lubricant after every 6 months or after 1000 hrs of operation and thoroughly flush the gear housing.	
			X			Check intake air filter for contamination, clean or replace as required.	
				X		Clean dirt from the housing	
		X				Clean fan cover, fan wheel, ventilation grille and cooling fins	

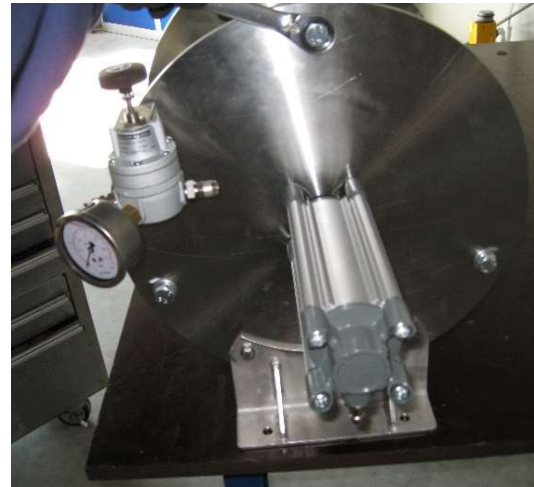
7.2.12 Maintenance & Cleaning of the Screen Basket



WARNING: Before starting, ensure the compressed air hose to the pneumatic cylinder is disconnected, the cylinder is de-pressurised and has been secured against reactivation.

Disassembly of Cylinder Retaining Plate

Loosen and remove the screws on the discharge chute retaining plate, remove the retaining plate including the air cylinder and press cone.



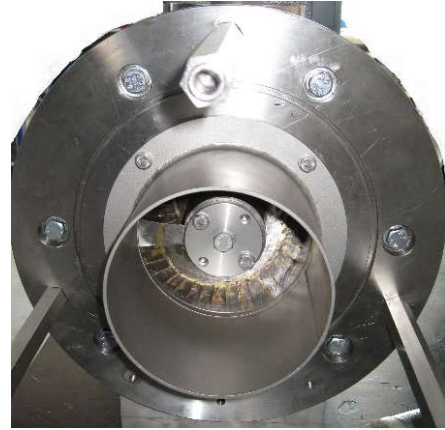
Removal of the Discharge Chute Retaining Plate and Enclosure

Use a mallet to loosen the retaining plate from the retaining rods.



Discharge Chute Removed

Please note that the housing is only clamped and that it will slip out of the guide grooves when you remove the cylinder retaining plate.



Removal of the Press Zone

Loosen the 4 screws and remove the press zone.



Removal of the Screen Basket

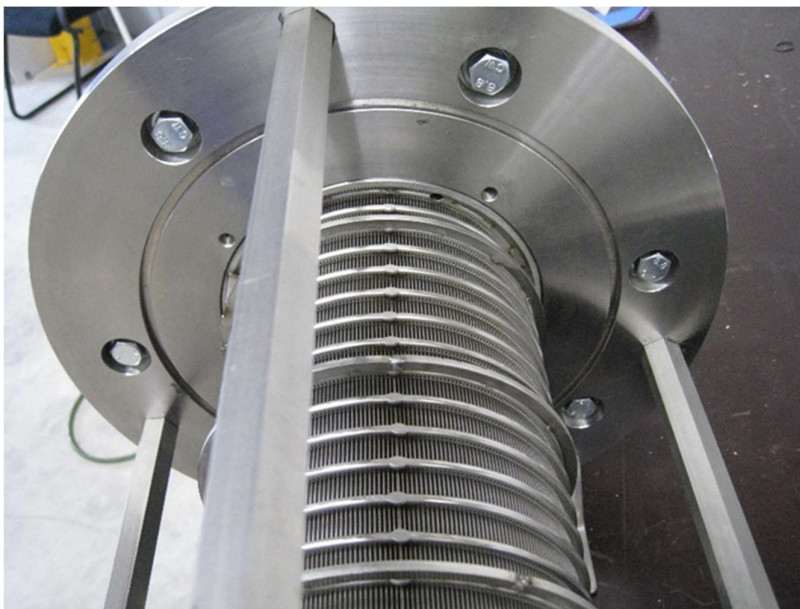
Now the screen basket can be pulled out of the separator using care.

Ensure that this is carried out as straight as possible to prevent the screen basket jamming!

Once the screen basket has been pulled out of the housing by roughly half of its length, the initial resistance experienced reduces considerably.



Take Note: Care must be taken that neither the screen basket nor the screw conveyor are damaged by the removal process. Particular care must be taken that the flange of the screen basket does not tilt when removing it, otherwise it may jam in place.



Picture: Removal of the Screen Basket

Check the Screen Basket

When the screen basket has been removed and carefully cleaned, prior to re-installation it should be checked to see if damage exists on the surface and that seamless movement of the screw is ensured. The screen basket should not have any deformations or oval distortions. If there is any uncertainty as to what distortion tolerance is permissible, contact FMD.

Any new screen should also be subjected to an inspection for deformations and distortion prior to being fitted.



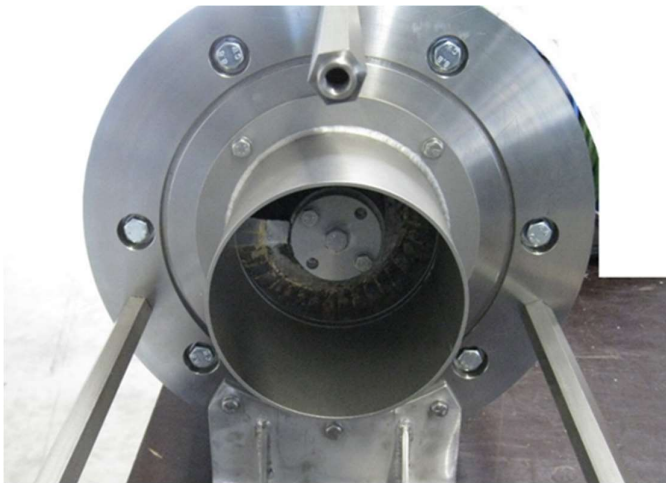
Picture: Screen Basket

Screen Basket Installation

The installation of a cleaned or new screen basket occurs in the reverse order to disassembly. When inserting the screen basket, be sure to push it in as straight as possible in order to prevent it jamming and potentially damaging the screw or basket.

Fitting the Press Zone

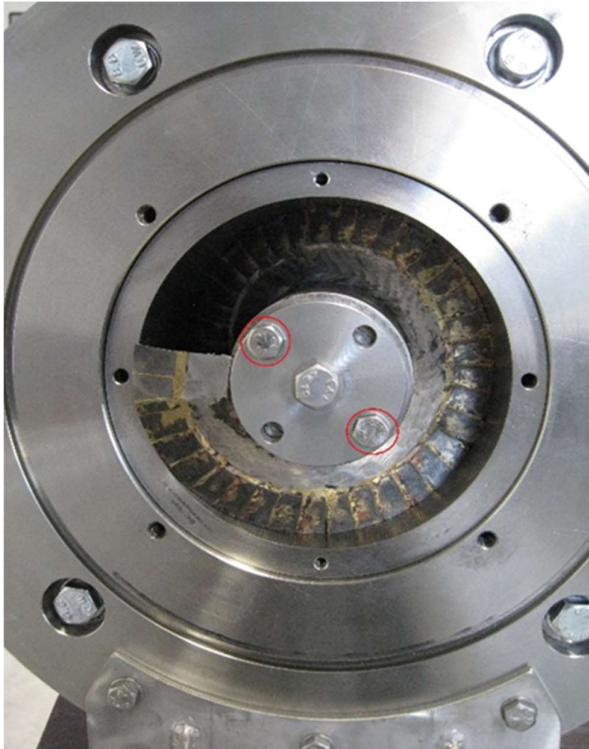
If the screen basket has been inserted correctly, then the press zone can be re-attached to the press. Ensure it easily fastens without any gap to the main housing.



After attaching the press zone, check all screws and tighten to the correct torque of 35 Nm.

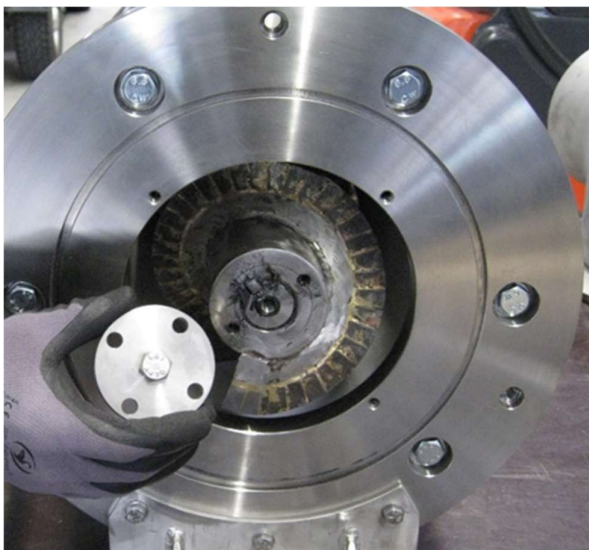
7.2.13 Maintenance of the Screw

If it is necessary to change the screw or carry out an inspection, then first remove the screen basket as described above. After dismantling the screen basket, the outer holding screws of the screw mounting plate can be released.



Screw Release (circled red)

Rotate the screw fixing plate in an anti-clockwise direction.



Removing the Fixing Plate

Pulling out the Screw

The locating holes for the screw holder can be used for inserting a tool for pulling the screw from the shaft (available on request from FMD).

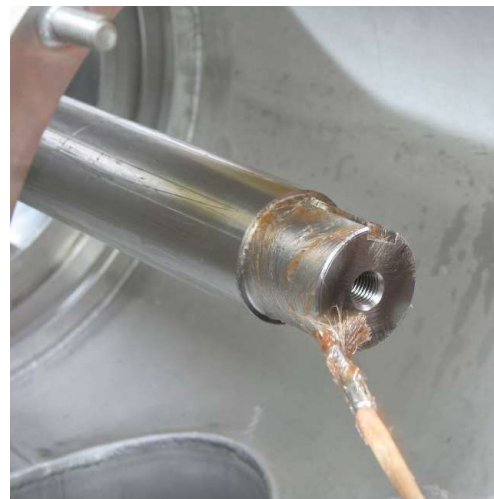
Once the screw has been released and the initial friction overcome, the screw can be pulled out carefully. Two people may be required to lift it out.



CAUTION: As the screw is pulled through the rear hole, **DO NOT** allow the screw to drop freely into the housing! The carbide edges of the screw are brittle and easily damaged by impact!

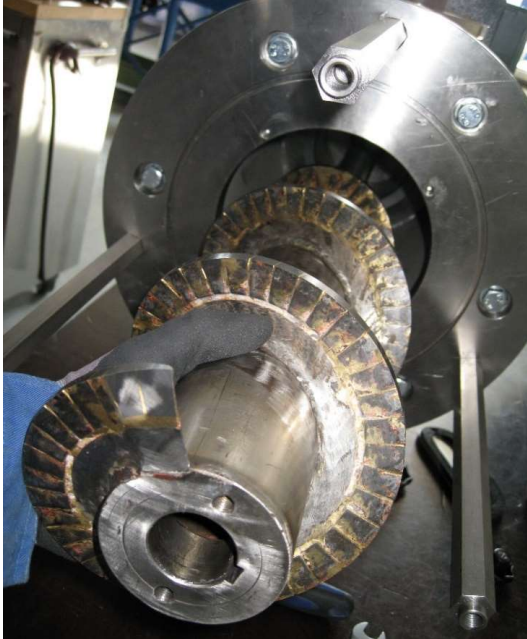
Greasing of Rear and Front

Prior to reinserting a renovated or new screw, the holding and mounting zones of the screw must be treated with grease so that installation (and subsequent removal) can be performed easily:



Inserting the Screw

Lift the screw onto the shaft and make sure not to strike the carbide edges of the screw as they are easily damaged by impact.



Screw in Fully Home Position

At the end of the mounting zone, the resistance when introducing the screw will be noticeably greater. At this point, the fully home position should be reached carefully using a soft-face hammer.

Attaching the Fixing Plate

Finally, the fixing plate must be attached to the shaft. In this process, the correct play-free installation must be ensured and the correct tightening torque applied.

After this step, the screen basket can be reinserted as described previously.

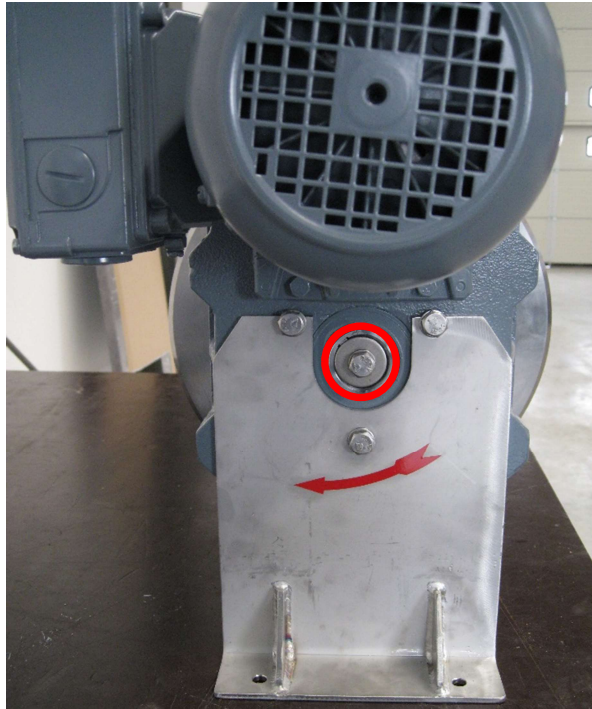


Picture: Re-attaching of the Fixing Plate

7.2.13.1 Maintenance of the Shaft & Shaft Seals

Loosening the Retaining Plate Screw

If the screen basket and the screw are removed, maintenance work and replacement of the shaft and the shaft seals can be performed. To do this, the screw of the retaining plate is to be released on the gear side.



Picture: Release the Retaining Plate Screw (red circle)

Driving out the Shaft

The shaft can be driven forward to the press side by lightly hitting it with a soft-faced hammer. Ensure that the shaft is not knocked out completely.



Picture: Drive the Shaft Forward

Loosening and Removing the Shaft

If the shaft has been loosened, it can be pulled out to the front/separator side. The feather key will fall out and remain on the gearbox side. Make sure that the feather key does not fall out of the housing. The key and shaft should be cleaned of dirt and grease residues before they are reinstalled.



Shaft seals in Position

During shaft removal, be sure not to damage the shaft seats or the holder for the shaft seals located in the rear housing.



Greasing the Holder

Once the shaft has been removed, the shaft seals located in the rear housing can be removed. After an inspection or replacement of the shaft, the shaft seals must always be replaced. The holder for the shaft and for the shaft seals must be cleaned and greased prior to inserting the shaft.



Feather Key Insertion

After cleaning and applying grease, the shaft can be reinserted. During this process, it must be ensured that the bracket for the feather key on the shaft matches the location on the gear unit. The feather key can be introduced through light hitting with a soft-faced hammer.



Mount the Retaining Plate

Once the feather key has been inserted flush, the retaining plate can be attached. For corrosion protection, the grease discharged during installation can be distributed on the surface and the excess grease removed.

The retaining plate must be attached with the specified torque.



Inserting the Rear Shaft Seal

Now the two shaft seals can be inserted, ensuring they are greased prior to insertion.



Inserting the Front Shaft Seal

If the rear shaft is inserted, then the front shaft can be inserted. It must be ensured that careful and correct positioning of the seal is carried out.



Correctly Inserted Shaft Seal

If both shaft seals have been inserted correctly, they seal flush with the housing. When the seals are properly inserted, the installation of the screw and the screen basket can be carried out as previously described.



7.3 Vacuum / Transfer Pump

7.3.1 Safety Precautions

This manual contains basic instructions which must be observed when installing, operating and servicing the vacuum / transfer pump. It is essential for the user / installer or responsible technician to read the manual thoroughly prior to installation and operation.

A copy of the manual must always be at hand where the pump is being used. In addition to the general safety instructions listed in the main manual section on safety, it is necessary to observe the specific safety rules included in the various sections of the manual.

It shall be the plant operator's responsibility to ensure that all maintenance, inspection and assembly work is performed by authorized and qualified personnel who have adequately familiarized themselves with the subject matter by studying this manual in detail.

Any work on the machine shall only be performed when it is at a standstill, it being imperative that the procedure for shutting down the machine described in this manual be followed. Pumps and pumps units which convey hazardous media must be decontaminated.

Replacement parts that have not been supplied by the manufacturer are not tested and not approved for use. The installation and usage of such replacement parts can have a negative impact upon the pump characteristics.

The manufacturer is not liable for any damages caused by the usage of non-original parts.

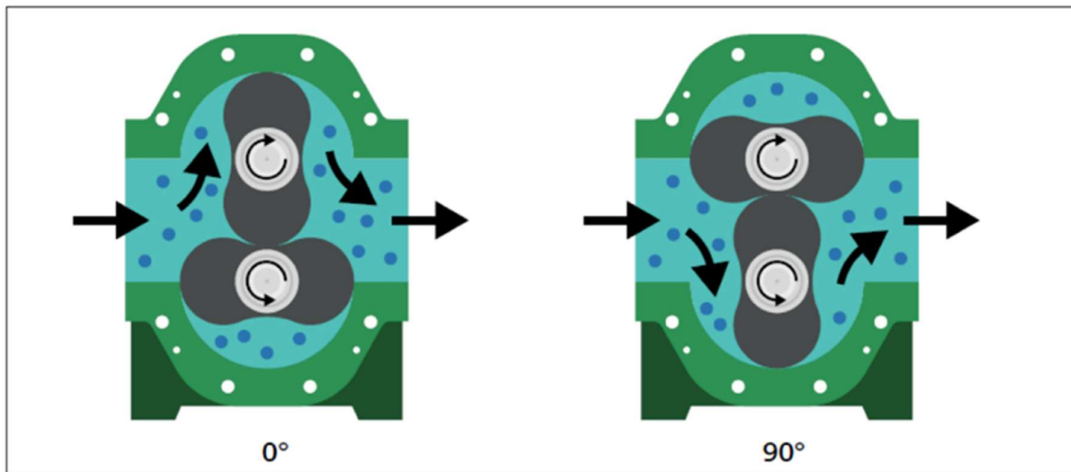
7.3.2 Technical Data

Pump Type	MB-2
Serial Number	D4093386
Power	3.0 kW
RPM	462rpm (at 50 Hz)
Flow Rate	16 m ³ /hr (at 50 Hz)
Max. DP	6 bar

7.3.3 Description

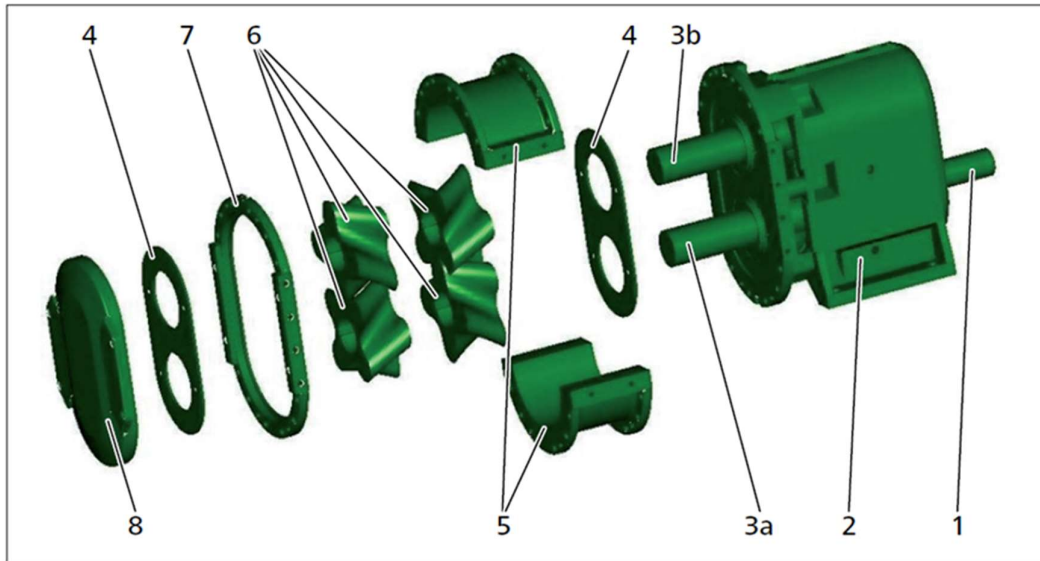
The illustrations in these instructions may not always correspond exactly to the installed machine. But they correctly describe the basic structure and the sequence of work steps.

The vacuum / transfer pump is a rotary lobe pump, which is of a positive displacement type. The main components which determine the pumping principle are two lobe rotors rotating in opposite directions within a housing, basically as shown below:



7.3.4 Main Parts

The main parts of the lobe pump are shown below:



(1)	drive shaft	(5)	hull
(2)	housing	(6)	lobe rotor
(3)	shaft	(7)	plate
(4)	wear plate	(8)	front cover

- The drive is attached to the drive shaft (1).
- The drive shaft (1) is the extension of one of the two shafts (3) of the housing (2).
- In the housing (2) the movement of the drive shaft (1) is transmitted to shaft (3b), which turns in the opposite direction.
- The pump housing is formed by the front plate of the housing (2), the two hulls (5), the two wear plates (4), the plate (7) and the cover plate (8).
- The lobe rotors (6) transport the medium within the pump housing.
- Different versions of this pump have different forms and numbers of lobe rotors (6).

7.3.5 Storage / Transport

The pumps are preserved for transport and short-term storage unless specified otherwise. In cases of longer storage, the pumps should be handled as follows until commissioning:

Storage of Lobe Rotors

The following applies to a storage period of up to six months. Standard DIN7716 summarizes detailed information on the storage of rubber products, of which the following is an extract:

General

Most rubber products may change their physical properties under unfavourable conditions or if treated improperly, which will result in a shorter lifetime. They can become unsuitable for use through excessive hardening, ageing, degeneration or permanent deformation, or because of blistering, cracking or other damage appearing on the surfaces.

These changes may occur under the influence of oxygen, ozone, heat, light, humidity, solvents or because of storing the products under load.

If stored and treated properly, rubber products will retain their properties even during long term storage (some years) with little change. However, this does not apply to uncured rubber compounds.

Storage Environment:

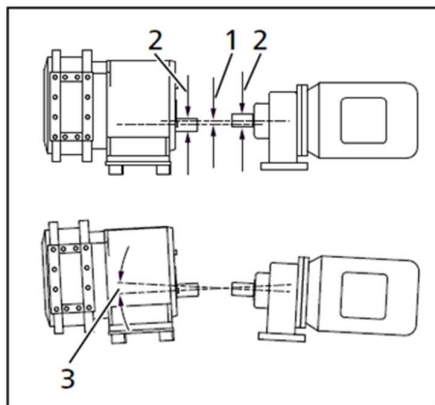
- *The environment in which rubber products are being kept must be cool, dry, free of dust and naturally ventilated.*
- *Rubber products must not be stored in the open, not even in an area sheltered from the weather.*
- *Rubber products should be stored at temperatures above minus 10 °C and not more than plus 15 °C. Storerooms should not be damp and there should be no condensation.*
- *Rubber products should be stored in an environment offering a relative humidity under 65 %.*
- *Rubber products must be protected against light, particularly direct sunlight or high UV artificial.*
- *Rubber products should be kept away from forced ventilation and draughts; under these conditions they should be wrapped.*
- *As ozone is very aggressive and harmful to rubber products, they should not be stored in the same room with equipment likely to produce ozone, e.g. electric motors or other equipment which could electrically discharge or cause sparks.*
- *Rubber products should not be stored together with solvents, oil, grease, lubricants or any chemicals.*

7.3.6 Installation Instructions

Unless specified otherwise the pump is designed for a horizontal mounting, bolted securely to a sub-structure. The vent plug of the housing must always be mounted at the top of the gear box.

7.3.6.1 Alignment of Pump and Drive

Pumps supplied as a machine complete with baseplate and drive will be aligned when assembled in the factory.



After bolting the baseplate to the foundation, remove the coupling guard and check the alignment of the coupling with a ruler and re-align if necessary. There is danger of distorting the baseplate if the foundation is uneven which will result in misalignment.

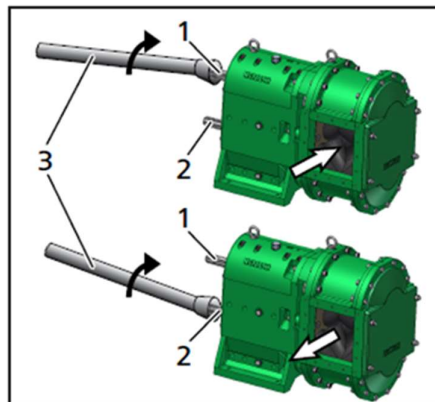
Permitted maximum misalignment tolerances for the coupling between drive and pump are:

- Radial shift (1): 1% of the largest diameter (2)
- Angular misalignment (3): 1°30'

No axial forces are allowed.

Check the alignment after a short test run and make corrections if necessary.

7.3.6.2 Direction of Rotation and Flow



The lobe pumps are reversible and therefore direction of rotation can be selected to provide for the required direction of flow.

There are two ways to change the direction of rotation/flow:

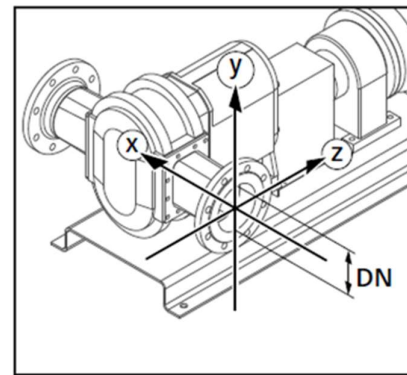
- Pumps with one drive shaft:
by changing the direction of rotation of the drive
- Pumps with two drive shafts:
by switching the cardan shaft (3) from drive shaft (1) to (2)

7.3.6.3 Piping System

- Clean the pipework and rinse thoroughly before installing the pump.
- Arrange the inlet and outlet pipework so that when the pump is not running, the pumped media is still present before and after the pump (pump remains flooded); sufficient pumped media should remain inside the pump head to lubricate the pump head components during restart.
- Connect the pipework ensuring sufficient pipe supports are used so that no external forces or stress act on the pump. Where required pulsation dampers or compensators can be used between the pump and pipework

The maximum allowable twisting loads (F_x , F_y , F_z) and bending loads (M_x , M_y , M_z) for the pump inlet and outlet connections have to comply with the requirements of API 676 and EN ISO 14847:

Standard nominal diameter [DN]	F_x , F_y , F_z [N]	M_x , M_y , M_z [Nm]
50	650	350
65	850	435
80	1040	560
100	1300	700
125	1700	865
150	1950	1050
200	2600	1400
250	3250	1750
300	3900	2100



7.3.6.4 Electrical Connection

Electric current can kill or cause serious injuries. All work relating to the electrical supply should be carried out by authorized and qualified personnel and should be in compliance with the requirements of the relevant national regulations.

In particular for control systems the latest version of the EC-Directive for Machinery, Annex I, Section 1.2 Controls should be observed.



WARNING: Follow all local and national electrical codes for all wiring. A qualified electrician must perform all wiring. Failure to comply may result in severe personal injury or property damage

7.3.7 Start-Up



Take Note: Never run the pump dry! - A few rotations in dry condition will damage the rotor lobes.

Pay strict attention to the following:

- Before starting up for the first time, fill the pump with medium to prime it.
- Priming is vital to initially lubricate the rubber lobe parts.
- Open valves and vents before starting the pump.
- Check the direction of rotation and flow by briefly switching on the pump drive.



CAUTION: This a positive displacement pump, which is capable of generating pressures that could cause damage – in extreme cases the bursting of pipes or vessels.



CAUTION: Possible damage to the pump by exceeding the pressure rating of the pump head and inlet and outlet connections. Never run the pump against a closed valve.

7.3.8 Temporary Shutdown

General

- Stop the pump by turning off the power and make sure it cannot be turned on accidentally.
- Drain the pump head and if necessary, wash through if there is a risk of freezing or if the medium may solidify.

Standby Pumps

Where a standby pump is installed as a back-up for the main pump and it is likely to stand idle for an extended period then it is recommended that it is operated from time to time.

7.3.9 Maintenance

7.3.9.1 Pump in General

The pump head should be regularly washed through or rinsed out if sediment from the pumped media is likely to settle out in the pump head.



WARNING: Hands can be crushed by moving parts! Before opening the pump head (removing the cover plate) turn off the power supply and ensure it cannot be turned on accidentally (remove fuse).

7.3.9.2 Lubricating the Shaft Seals

The pump shaft seals are lubricated with grease via two grease points on the side of the gear housing, with two grease outlet points on the opposite side.

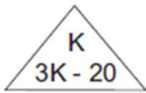
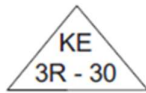
Lubrication should always be carried out while the pump is running – **never during standstill**. For this purpose, a high duty water-resistant grease, type NLGI – class 2, should be applied.

Lubrication Schedule

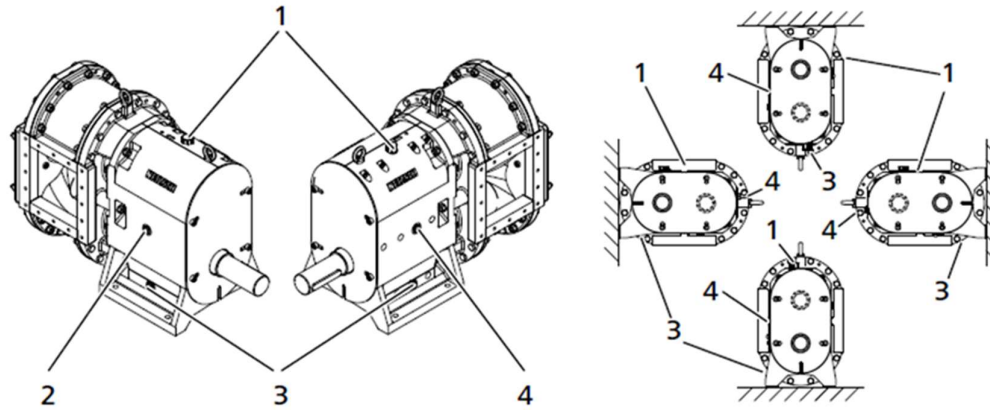
- After every 350 hours of operation

NB: The filling quantity with the hand-lever grease gun should not exceed 2 to 3 strokes per grease nipple.

Recommended Lubricating Grease:

Temperature range of the grease [°C]	Designation DIN 51502	Example
-30 ... +120		"SHELL" Alvania G3
-30 ... +180		"KLÜBER" PETAMO GHY 443

7.3.9.3 Oil Filling & Oil Changing of the Housing



The deairation screw (1) has always to be at the top position.
The magnetic oil drain screw (3) has always to be on the lowest position.

Draining

- Open the magnetic drain screw (3) and let all the oil flow out.
- Clean the magnetic oil drain screw (3) at every oil change.
- Close the magnetic oil drain screw (3).

Filling

- Fill through the deairation screw (1) until approx. the middle of the sight glass (2).
- Close the deairation screw (1).

7.3.9.4 Oil Quantities & Specification

The housing of the pump requires the following oil quantity:

Rotary Lobe Pump Type	Oil quantity horizontal	Oil quantity vertical	Change interval*
XB	0.6L	0.91L	First change of oil after approx. 500 hours of operation or 12 months Then after each 2000 hours of operation or 12 months

Oil specification: SAE 75W-90 API GL-4 (or API GL-5)

* Change intervals are the base to maintain warranty; in aggressive conditions (high/low temp and humidity, full load operation etc.) the change interval can be shortened significantly.

7.3.10 Trouble-Shooting

The troubleshooting charts list possible problems, the possible cause and the potential remedy. For problems not listed FMD should be consulted.

Possible Problems											Possible Causes	Potential Remedy
The pump will not start.	The pump will not prime.	The pumped media flowrate is too low.	The generated differential pressure is too low.	The pumped media flowrate is unstable.	The pump is noisy.	The pump is stalled.	The drive is overloaded.	The lobe rotors are worn at an early stage.	The housing is worn at an early stage.	The shaft seal is leaking.		
X							X				In new pumps or with new lobe rotors: the static friction is too great.	Fill the pump up, then pump through manually using a suitable appliance; if necessary use glycerine as lubricant in the pump housing.
X	X	X					X				The pump electrical equipment is not compatible with the electrical supply.	Check order documentation. Check electrical supply/installation (possibly 2 phase operation).
		X					X	X	X		The discharge/differential pressure is too high.	Measure the pressure and check against order documentation. Reduce pressure or change the drive. Measure the pressure and check against order documentation. Reduce pressure or change the drive.
X						X	X				There are foreign bodies in the pump.	Remove foreign bodies and check lobe rotors and housing for possible damage; replace if necessary.
X						X	X	X	X		The temperature of the pumped media is too high, the expansion of the lobe rotors is too high.	Lower the pumped media temperature.
X						X	X	X			The lobe rotors have swollen, the elastomer is not compatible with the medium.	Check that the pumped media agrees with the order documentation. Possibly change the lobe rotor material.
X						X	X	X			The solids content of the pumped media causing a blockage.	Reduce the solids content of the pumped media.
X						X	X	X	X	X	The pumped media settles or hardens when left to stand.	Clean out the pump head and rinse through after each use.
		X	X	X							There is air in the inlet pipe.	Increase the inlet (suction) head (liquid level), tighten all connections in the inlet line, prevent turbulence and air bubbles at the inlet connection.
	X	X	X	X							The inlet pipe is leaking.	Check and tighten all inlet line connections to ensure they are airtight.
	X	X	X	X							The shaft seal is leaking.	Mechanical seal: renew damaged seal components/seals, clean out and eliminate deposits of solids.
		X	X								The pump speed is too low.	In the case of a variable speed drive: increase the speed. If necessary change the drive.

Possible Problems												Possible Causes	Potential Remedy
The pump will not start.	The pump will not prime.	The pumped media flowrate is too low.	The generated differential pressure is too low.	The pumped media flowrate is unstable.	The pump is noisy.	The pump is stalled.	The drive is overloaded.	The lobe rotors are worn at an early stage.	The housing is worn at an early stage.	The shaft seal is leaking.	Sleeve (5360) is turning unrythmically or not at all.		
		X	X	X	X			X	X	X		The inlet (suction) pressure is too low/suction pressure is too high (cavitation).	Increase inlet pressure i.e.: lower the temperature of the pumped media, install the pump at a lower location, increase inlet line size and/or reduce length.
		X	X		X	X	X	X			X	The pump is running dry.	Prime the pump head with liquid, provide for dry running protection, rearrange the inlet pipework.
	X	X	X	X	X							The lobe rotors are worn.	Replace with new lobe rotors.
	X	X	X		X			X	X			The lobe rotor material is not compatible with the pumped media/pump duty conditions.	Fit new lobe rotors. Check the pumped media agrees with order documentation; if necessary change the lobe rotor material.
	X	X	X	X	X			X		X		Hulls/wear plates are worn.	Change housing and/or wear plates. Establish the cause (i.e. wear and tear, corrosion, cavitation etc.).
					X							The pump and drive are not correctly aligned.	Re-align the pump/drive (see section 4.2).
					X							The flexible element of the coupling is worn.	Replace the flexible element and re-align pump/drive (see section 4.2).
					X	X	X				X	The pump gearbox bearings are worn.	Replace bearings. Monitor and maintain the level of lubricant.
					X		X					The pump speed is too high.	In the case of a variable speed drive: reduce the speed. If necessary change the drive.
	X	X						X			X	The viscosity is too high.	Measure the viscosity and compare with order documentation. If necessary/possible adjust viscosity or change the drive.
	X	X			X		X					The specific weight of the pumped media is too high.	Measure specific weight and compare with order documentation. If necessary/possible adjust specific weight or change the drive.
											X	Mechanical seal is damaged: mechanical seal faces have failed.	Replace seal rings.
	X	X									X	Mechanical seal is damaged: elastomers have failed (swollen or brittle).	Replace elastomers. Check that the pumped media agrees with order documentation, if necessary change elastomer material.
											X	Lock nut (5450) is loose.	Tighten lock nut (see section 7.2)

7.3.11 Disassembly & Assembly of the Lobe Rotors

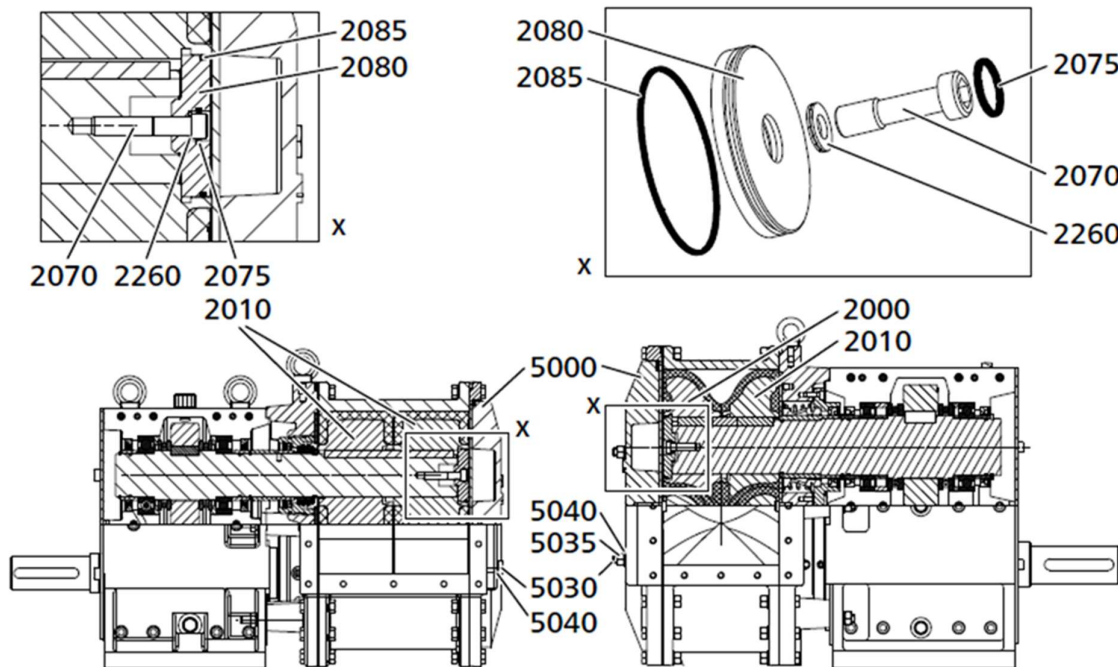
7.3.11.1 Disassembly



BIOHAZARD: Risk of illness and disease from the pumped medium! Decontaminate and empty the pump before opening. Wear appropriate personal protection equipment.



DANGER: Hands can be crushed by moving parts! Before opening the pump head (removing the cover plate) turn off the power supply and ensure it cannot be turned on accidentally (remove fuse).



- Remove screws/spring washers (5030/5040) or hex nuts/spring washers (5035/5040).
- Remove the cover plate (5000).
- Remove screws (2070 with O-ring (2075) and fuse (2260).
- Remove end parts (2080) with O-rings (2085).
- Pull off the lobe rotors (2010) **separately** using a lobe extractor (pull-off device).

7.3.11.2 Assembly

Assembly is a reversal of the above disassembly procedure, but additionally:

- In case of visible wear of the housings, crescents or wear plates these should be replaced.
- Fasten screws/nuts with a torque wrench as given in the following table:

Size	Material		Pos. no.	Remarks
	Stainless steel A...-70	Steel 10.9		
M8	30 Nm	40 Nm	2070	-
M10	40 Nm	50 Nm	2070	-
M12	60 Nm	90 Nm	2070	-
M16	150 Nm	200 Nm	2070	-
M24	100 Nm	120 Nm	2080	Types MB only
M45 x 1.5	100 Nm	120 Nm	5450	Types XLB-.../2 only

- Secure screw (2070) with Loctite 245 or similar
- Secure or seal other screws with material according to the section drawing

Technical Drawing Details:

- Top View (Cross-section):** Shows a complex assembly with various components labeled with part numbers: 0020, 0234, 0390, 0232, 0130, 0370, 0410, 0115, 0110, 0230, 0320, 0120, 0180, 0210, 2140, 2100, 2020, 2010, 5010, 2085, 2030, 2080, 2110, 2120, 5200, 5030, 5040, 5000, 2060, 5050, 2160, 2040, 2050, 2150, 2180, 2190, 0365. A note indicates: "mit Loctite 245 gesichert / secured with Loctite 245".
- Bottom View (Cross-section):** Labeled with 'X', showing components: 7080, 7030, 7010, 2105, 7070, 7040, 7000, 7050.
- Isometric View (Left):** Labeled with 'X', showing components: 7131, 7130, 7135, 2130, 0315, 2090, 0300, 0305, 0316, 0010. A note indicates: "mit Loctite 511 gesichert / secured with Loctite 511".
- Isometric View (Right):** Shows components: 7131, 7135, 7132, 0370, 7375, 7380, 7390, 0290, 7370, 7280.
- Assembly Instructions:**
 - opt. Ölquench / opt. oil quench
 - mit Loctite 511 gesichert / secured with Loctite 511
 - mit Loctite 245 gesichert / secured with Loctite 245
- Table:**

Schnitt-Zeichnung-Nr. / Drawing No.		Datum / date		Name		Unterschrift / Signature	
T-P021.000		22.02.2008		Colantu		Revision 5	
1: X		22.02.2008		Rosam		Erst für Ausgabe vom: / First for issue of:	
Abt./Storage		22.02.2008		Dank		Ersetzt für Version of: / Replaces for version of:	
		Freigegeben / released		22.02.2008		13.12.2005	

SPARE PARTS LIST				
Part Pos.	Qty	Unit	Designation Size / Dimension, Material / Surface	Part No
0010	1	Pce	housing	
0020	4	Pce	pin	
0110	1	Pce	drive shaft	
0115	1	Pce	drive shaft	
0120	2	Pce	wheel	
0130	1	Pce	top cover	
0180	2	Pce	bearing	
0210	3	Pce	lip seal	
0230	4	Pce	retaining ring	
0232	4	Pce	retaining ring	
0234	4	Pce	retaining ring	
0290	1	Pce	drain plug	
0300	1	Pce	sight glass	
0305	2	Pce	drain plug	
0310	1	Pce	screw	
0315	1	Pce	screw	
0316	3	Pce	gasket	
0320	2	Pce	key	
0365	10	Pce	cylindrical head screw	
0370	4	Pce	hex head screw	
0390	2	Pce	bearing	
0400	1	Pce	key	
0410	1	Pce	closure	
0500	0	L	lubricant oil	
2010	2	Pce	lobe rotor	
2020	2	Pce	hull	
2030	2	Pce	key	
2040	16	Pce	spring washer	
2050	16	Pce	hex nut	
2060	16	Pce	hex head screw	
2080	2	Pce	end part	
2085	2	Pce	O-ring	
2090	8	Pce	pin	
2100	2	Pce	shaft	
2105	2	Pce	O-ring	
2110	2	Pce	cylindrical head screw	
2120	1	Pce	wear plate	
2130	4	Pce	countersunk screw	
2140	1	Pce	wear plate	
2150	1	Pce	intermediate plate	
2160	4	Pce	flat seal	
2170	6	Pce	stud screw	
2180	6	Pce	spring washer	
2190	6	Pce	hex nut	

SPARE PARTS LIST				
Part Pos.	Qty	Unit	Designation Size / Dimension, Material / Surface	Part No
5000	1	Pce	pump cover	
5010	1	Pce	seal	
5030	4	Pce	hex head screw	
5040	4	Pce	spring washer	
5050	1	Pce	plate	
5200	2	Pce	pin	
7000	2	Pce	seal housing	
7010	2	Pce	top cover	
7030	2	Pce	lip seal	
7040	4	Pce	O-ring	
7050	2	Pce	mechanical seal	
7070	2	Pce	fuse	
7080	2	Pce	retaining ring	
7130	2	Pce	nipple	
7131	4	Pce	reduction piece	
7135	2	Pce	valve	
9020	1	Pce	coupling part	
9025	1	Pce	coupling part	
9030	1	Pce	gear rim	

7.4 Liquor Transfer Pump



7.4.1 Safety Precautions & Symbols



Take Note: Before using the pump carefully read the information contained in this instruction manual.



WARNING: Electrical Hazard! ALWAYS check for no voltage before starting work!



WARNING: Electrical connection must be carried out by suitably a qualified electrician.

7.4.2 Technical Data

Nominal Speed	2900/3450 rpm
Protection	IP 54
Supply Voltage / Frequency	480V 3~ 50/60 Hz

7.4.3 Description

Close-coupled, centrifugal pumps; electric motor with extended shaft directly connected to the pump with open impeller.

Intended use for moderately dirty liquids (maximum size of solids: 4 mm), non-explosive, non-aggressive for the pump materials. Liquid temperature from -10 °C to +90 °C.

7.4.4 Start-up and Operation



*Take Note: When first starting, check the direction of rotation is as per the arrow on the casing
Never run the pump dry, only start the pump after filling with liquid and venting the case.*

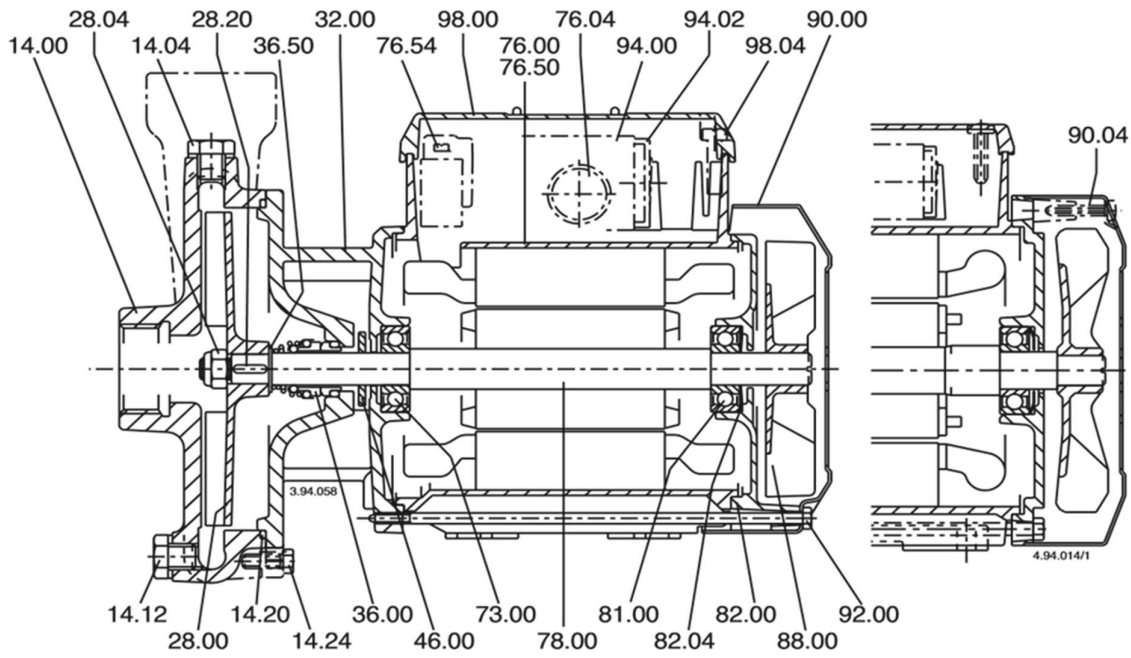
Pay strict attention to the following:

- Open valves and vents before starting the pump.
- Before starting up for the first time, fill the pump and vent the casing using the vent fitted.
- Check the direction of rotation and flow by briefly switching on the pump drive.
- If the pump discharge piping is routed above the liquor tank a non return valve will be required to prevent back filling of the food waste system, a valve suitable for water with sediment should be selected.

7.4.5 Troubleshooting

PROBLEM	PROBABLE CAUSES	POSSIBLE REMEDIES
(1) The motor does not start	1a) Motor overload protection cuts in 1b) Shaft locked	1a) Check the power supply and make sure that the pump shaft is turning freely. Check that the thermal overload protection has been set correctly. 1b) Remove the cause of lockage.
(2) Pump locked	2a) Prolonged periods of inactivity forms rust inside the pump 2b) Presence of solids in impeller 2c) Bearings seized	2a) Free the pump by rotating the shaft manually. 2b) Remove the wet end casing and remove any solid solids the impeller. 2c) Replace the bearings.
(3) Pump runs with no discharge	3a) Possible air leak on suction. 3b) Suction is blocked.	3a) Check. 3b) Clean blockage from suction.
(4) Low flow discharge.	4a) Presence of deposits or solid on the impeller 4b) Impeller damaged or worn 4c) Worn pump case	4a) Clean the impeller. 4b) Replace the rotor. 4c) Replace the pump casing.
(5) Noise and vibrations from the pump	5a) Rotating part unbalanced 5b) Worn bearings	5a) Check that no solid bodies are obstructing the rotor. 5b) Replace the bearings.
(6) Leakage from the mechanical seal	6a) The mechanical seal has been run dry or has stuck 6b) Mechanical is damaged 6c) Slight initial drip during filling or on first start-up.	6a) Replace the mechanical seal. 6b) Replace the mechanical seal. 6c) Replace the mechanical seal.

7.4.6 Spare Parts List



SPARE PART NUMBERS			
14.00	Pump Casing	70.00	Lantern bracket
14.04	Plug (filling)	70.18	Screw
14.12	Plug (draining)	70.19	Nut
14.20	O-ring	70.20	Screw
14.24	Screw	73.00	Pump side bearing
16.00	Suction casing	76.00	Motor casing with winding
16.04	Plug with washer	76.04	Cable gland
16.12	Air release plug with washer	76.16	Support
16.20	Casing gasket	76.20	Pin
18.00	Interstage plate	76.54	Terminal box set
20.00	Delivery casing	78.00	Shaft with rotor packet
20.04	Plug with washer	81.00	Fan side bearing
20.12	Screw	82.00	Motor end shield, fan side
20.00	Impeller	82.04	Compensating spring
28.04	Impeller nut	88.00	Motor fan
28.12	Circlip	90.00	Fan cover
28.20	Impeller key	90.04	Screw
30.00	Impeller	90.00	Fan cover
32.00	Lantern bracket	90.04	Screw
32.30	Guard	92.00	Tie bolt
32.32	Screw	94.00	Capacitor
32.32	Caged nut	94.02	Capacitor gland
36.00	Mechanical seal	98.00	Terminal box cover
36.50	Shoulder ring	98.04	Screw
46.00	Deflector	98.08	Gasket

7.5 Actuated Valves

7.5.1 Safety Precautions



Take Note: Ensure you have read the safety definitions and understand all instructions before servicing this machine. Failure to do so could result in personal injury or property damage.



CAUTION: Before working on a valve ensure system pressure has been released and pipes drained.



CAUTION: When installing or maintaining the actuated valve shut off and lock out power before removing any covers. Never make adjustments while in operation, except for those prescribed in this manual.



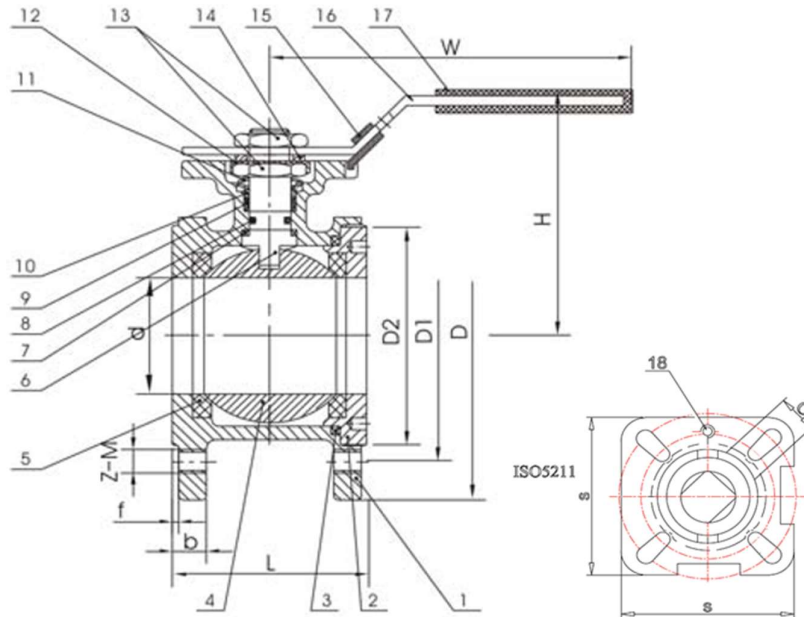
WARNING: Electrical Hazard! ALWAYS check for no voltage before starting work!

7.5.2 Valve Description

The stainless steel one-piece flanged ball valve with full bore is designed for service with vacuum and up to nominal pressure at a max 80degC. with ISO5211 mounting pad for electric actuation.



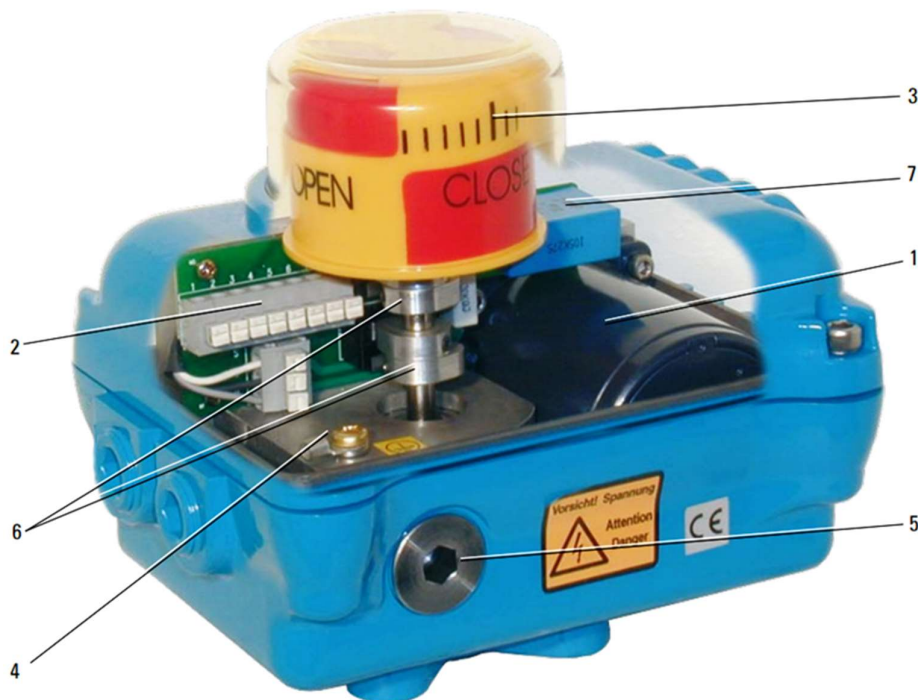
7.5.3 Exploded View



Pos.	Description	Material
1	Body	Stainless steel 1.4408
2	End Cap	Stainless steel 1.4408
3	Body seals	PTFE
4	Ball	Stainless steel 1.4401
5	Ball seats	PTFE + 15% GFR
6	Stem	Stainless steel 1.4401
7	Thrust washer	PTFE
8	O-ring	Viton
9	Stem seals	PTFE
10	Gland ring	Stainless steel AISI 304
11	Spring washer	Stainless steel AISI 304
12	Locking tab	Stainless steel AISI 304
13	Lever nut	Stainless steel AISI 304
14	Plate washer	Stainless steel AISI 304
15	Lock device	Stainless steel AISI 304
16/17	Handle	Stainless steel AISI 304
18	Stop pin	Stainless steel AISI 304

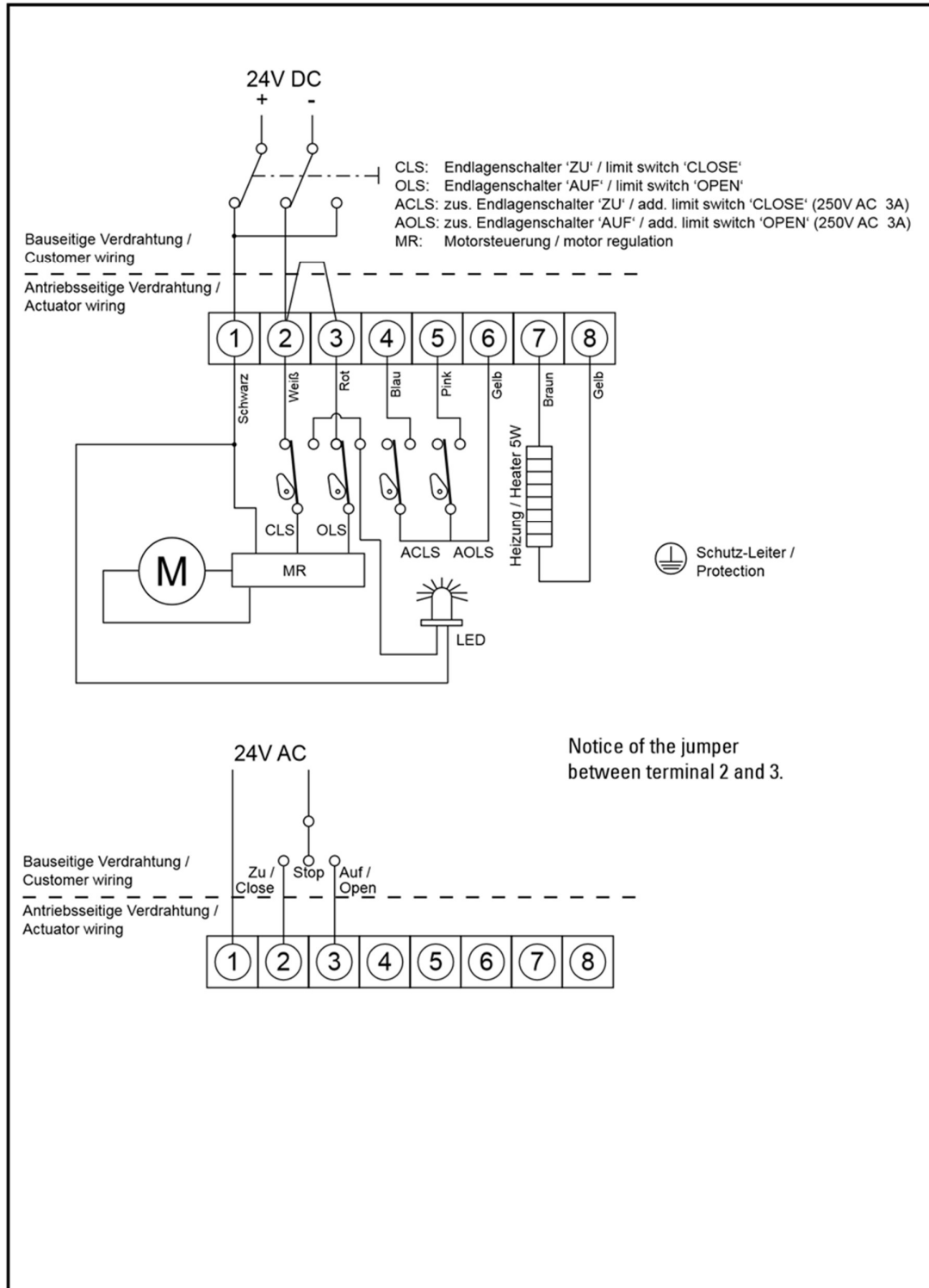
7.5.4 Actuator Description

The actuators are suited to the operation of ball and butterfly valves with a 90° rotation angle. They feature a water-resistant body with overload protection and a heating device against condensation formation. Power is transferred to the driveshaft via a self-stopping reduction worm gear to achieve precision control.



Pos.	Description
1	Motor
2	Terminal block
3	Valve position indicator
4	Gearbox
5	Manual override
6	Limit switches
7	Heater

7.5.5 Wiring Diagram



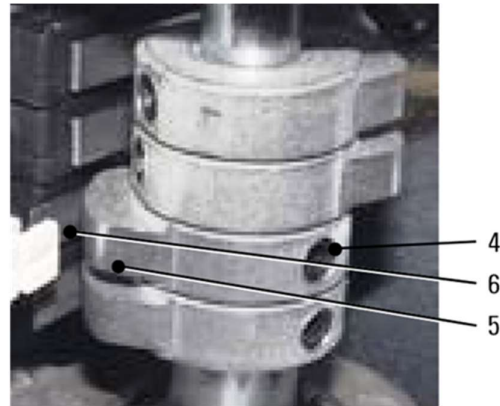
7.5.6 Adjustment of the Limit Switches

Adjustment of 1st "CLOSED" Limit Switch

Manually set the valve in the closed position. Loosen screw (1) on the lower cam (2) and adjust the cam to contact the lower micro switch (3). If the mechanical stop prevents the cam rotating sufficiently turn the mechanical stop screw 2 turns counterclockwise from the closed position and lock it off. After adjustment of the cam (2) secure the cam with the screw (1).

Adjustment of 2nd "CLOSED" Limit Switch

Loosen screw (4) on cam (5) and adjust the cam to contact the micro switch (6). After adjustment of cam (5) secure the cam with the screw (4).

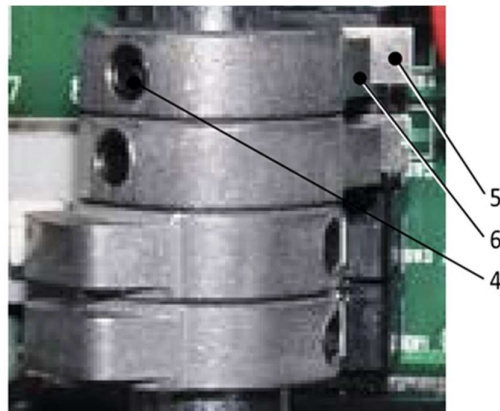
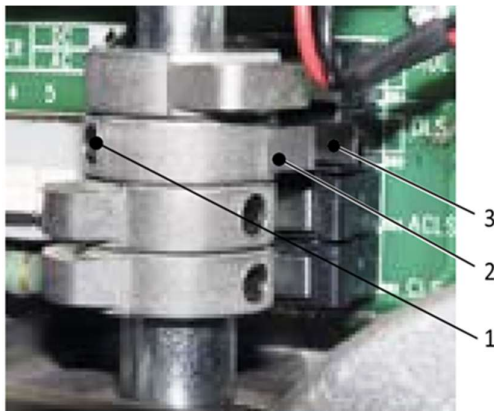


Adjustment of 1st "OPEN" Limit Switch

Manually set the valve in the open position. Loosen screw (1) on the upper cam (2) and adjust the cam to contact the upper micro switch (3). If the mechanical stop prevents the cam rotating sufficiently turn the mechanical stop screw 2 turns counterclockwise from the closed position and lock it off.

Adjustment of 2nd "OPEN" Limit Switch

Loosen screw (4) on cam (5) and adjust the cam to contact the micro switch (6). After adjustment of cam (5) secure the cam with the screw (4).



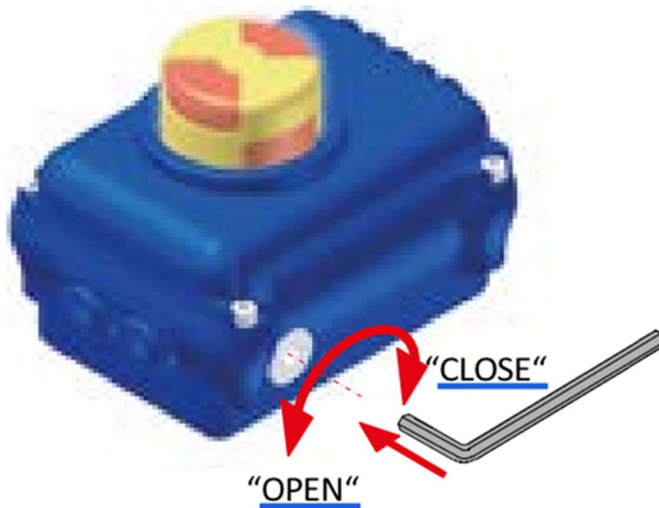
7.5.7 Emergency Operation

In the case of a power or control failure, or a fault in the actuator, the actuator has the capability for manual operation. Take care about the turning direction of the manual override.

The actuator is fitted with an Allen key on the underside, it fits in the socket on the end of the gear shaft. To close the actuator, turn the Allen key clockwise and to open the actuator turn the Allen key counterclockwise.



CAUTION: When finished with manual operation, remove the Allen key from the gear shaft to prevent the injury of persons or the damage of the actuator by normal operation of the electric actuator.



7.5.8 Valve List

P&ID Pos Nr.	Service Function	Type	Part No.
V.00.01.01	Dry Running Protection Solenoid	G1/2" Solenoid, 2/2-way, 24Vdc	A00103
V.00.02.03	Discharge Overboard / Ashore	Blank Flange Fitted	
V.00.03.01	Food waste Macerator Suction Valve	DN50 Ball Valve, Actuated 24Vdc	A00181
V.00.04.01	Hot water Flush	G1/2" Solenoid, 2/2-way, 24Vdc	A00103
V.00.05.01	Compressed Air to Press	G1/4" Solenoid, 2/2-way, 24Vdc	A00085
V.00.06.01	Pump Discharge to Press	DN50 Ball Valve, Actuated 24Vdc	A00181
V.00.07.01	Food waste Tank Suction	DN50 Ball Valve, Actuated 24Vdc	A00181
V.00.08.01	Pump Discharge to Press	DN50 Ball Valve, Actuated 24Vdc	A00181
V.00.09.01	Press inlet isolation valve	DN50 Ball Valve, Manual	A00094
V.00.10.01	FW Liquor Transfer Pump Suction	G1 1/2" Ball valve, Manual	A00019
V.00.11.01	FW Liquor Transfer Pump Discharge	G1 1/2" Ball valve, Manual	A00019
V.01.01.01	Macerator 1 Discharge Valve	G2" Ball Valve, Actuated 24Vdc	A00028
V.01.02.01	Macerator 2 Discharge Valve	G2" Ball Valve, Actuated 24Vdc	A00028
V.02.01.01	Macerator 1 Transport Water Valve	Solenoid, 2/2-way, SS/NBR, 24Vdc	A00029
V.02.02.01	Macerator 1 Transport Water Valve	Solenoid, 2/2-way, SS/NBR, 24Vdc	A00029

7.6 Sensor List

Refer to Annex 2 for P&ID-200429 showing sensor locations:

Refer to Annex 3 for sensor data sheets:

P&ID Pos Nr.	Service Function	Type	Part No.
M.00.01.01	Tank level HHL	Fixed point sensor, LMT100	A00071
M.00.02.01	Tank level	4-20mA Level sensor, PI2898	A00068
M.00.03.01	Pump suction / discharge pressure	4-20mA pressure/temp PM1504	A00152
M.00.04.01	Pump suction / discharge pressure	4-20mA pressure/temp PM1504	A00152
M.00.05.01	Press Feed Pressure	4-20mA pressure PI2898	A00068
M.00.06.01	Press Air Pressure Loss	4-20mA pressure PV7004	A00030
M.00.07.01	Bag Full Sensor	Laser distance OGD580	A00172
M.00.08.01	FW Liquor Transfer Tank Level	4-20mA Radar level LR9020	A00031
M.01.01.01	Macerator 1 Lid Position Switch	Position switch, FA 4510-2DN	
M.02.01.01	Macerator 2 Lid Position Switch	Position switch, FA 4510-2DN	

8 Annexes

8.1 Annex 1 Drawings

8.2 Annex 2 Sensors